NeuroPAC: A Pre-Processing, Analysis, and Classification Pipeline Platform for EEG Signals

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A Thesis Submitted in Partial Fulfillment of the Requirements

for the Degree of Bachelor of Science in Software Engineering

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Abstract

NeuroPAC is a user-friendly Graphical User Interface (GUI) designed to streamline the preprocessing, analysis, and classification of Electroencephalography (EEG) data. Built on top of the powerful FieldTrip toolbox, NeuroPAC simplifies complex EEG data workflows, making advanced analysis accessible to researchers with varying levels of programming expertise. The platform integrates essential preprocessing steps, including filtering, artifact removal via Independent Component Analysis (ICA), and event decomposition. It offers a suite of analysis techniques such as Event-Related Potential (ERP), time-frequency, spectral, channel-wise connectivity, and inter-trial coherence analyses. Additionally, NeuroPAC incorporates a deep learning classification module utilizing the EEGNet, EEG-Inception and Riemannian Geometry architectures to classify EEG signals effectively. This thesis details the design and implementation of NeuroPAC, demonstrating its capabilities through application to clinical datasets, specifically focusing on Parkinson's Disease classification. The results highlight NeuroPAC's potential to facilitate reproducible and efficient EEG data analysis in both research and clinical settings.

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Chapter I

Introduction

I.1 Background and Motivation

Electroencephalography (EEG) is a non-invasive technique used to record electrical activity in the brain. It has become an essential tool in neuroscience research and clinical diagnostics due to its high temporal resolution, non-invasive nature, and relatively low cost. EEG signals are widely used to study brain function, cognitive processes, and neurological disorders such as epilepsy, Parkinson's disease, and Alzheimer's disease. EEG can help to diagnose certain conditions such as Alzheimer's disease by identifying characteristic patterns of brain activity associated with the disease. For example, EEG can detect changes in brain wave patterns that are indicative of cognitive decline, memory impairment, and brain disconnectivity. Additionally, EEG can be used to monitor the progression of neurodegenerative diseases and evaluate the effectiveness of treatments. By analyzing EEG data, researchers and clinicians can gain insights into the underlying mechanisms of neurodegenerative diseases and develop more effective diagnostic and treatment strategies.

3-Auditory Oddball Task is a widely used experimental paradigm in cognitive neuroscience to investigate attention, perception, and cognitive processing. In this task, participants are presented with a series of auditory stimuli, typically consisting of frequent standard tones, infrequent target tones and novel tones. The participants are instructed to respond to the target tones while ignoring the standard tones and aren't instruced on novel tones because they are unexpected and meant to capture attention involuntarily. The oddball task is designed to elicit specific event-related potentials (ERPs) in the EEG signal, such as the P300 component,

which is associated with attentional processes and cognitive evaluation of stimuli. The oddball task has been extensively used to study various cognitive functions, including attention, memory, and decision-making. It has also been employed in clinical research to investigate cognitive deficits in neurological and psychiatric disorders, such as schizophrenia, ADHD, and Alzheimer's disease.

I.2 Problem Statement

The core difficulty in Electroencephalography (EEG) data analysis lies in the trade-off between power/flexibility and usability/accessibility. Here are some limitations of two existing and most popular state of the art tools that are used for EEG pre-processing and analysis:

- **Fieldtrip, Complexity of Code-Based Toolboxes:** A high-performance, flexible toolbox like FieldTrip is essential for advanced analysis but rely on MATLAB code and scripting. This necessitates a high level of programming expertise, creating a significant barrier to entry and slowing down the research process for those whose primary focus is neuroscience or clinical application rather than coding and data structuring.

- **EEGLAB, Existing GUI Complexity:** While EEGLAB offers a Graphical User Interface (GUI), this interface can present a dense, overwhelming array of parameters, menus, and advanced options to a novice. This complexity often fails to make the fundamental, necessary workflow—preprocessing, processing, and classification—feel less intimidating or more welcoming to beginners.

There is a clear and persistent need for a dedicated, simplified interface that explicitly targets the essential and initial steps of EEG analysis, thereby democratizing the process.

NeuroPAC addresses the need for a streamlined, intuitive Graphical User Interface (GUI) designed to guide users through the crucial, initial steps of preprocessing, processing, and deep learning classification in a less intimidating manner. By building on the powerful computational back-end of toolboxes like FieldTrip, this GUI will abstract away the code, presenting only the core configuration options needed for a reproducible, foundational analysis. The efficacy of

this simplified pipeline must be demonstrated in a real-world, high-stakes context. There is a need to validate this new streamlined analysis tool by successfully applying it to challenging neurological datasets, specifically in generating robust and clinically meaningful deep learning classification results for conditions such as Parkinson's Disease.

In summary, the basic difficulty is the lack of a user-friendly tool that simplifies the essential EEG analysis pipeline (preprocessing, feature extraction analysis, and classification). Existing tools are either highly code-dependent (FieldTrip) or still too complex for beginners (EEGLAB). The perceived need is thus the development of a simplified, welcoming user interface that leverages existing powerful toolboxes to make foundational EEG analysis accessible, efficient, and reproducible, with its utility confirmed through successful application and validation in classifying clinical cohorts like Parkinson's Disease patients.

I.3 Purpose of the Study

The purpose of this study is two-fold: to address the usability gap in existing neuroscientific software and to validate the resulting platform through compelling clinical application.

- **Engineering Goal (Product):** To develop a user-centric Graphical User Interface (GUI) using Python that provides a streamlined, accessible, and intuitive workflow for the fundamental steps of EEG data analysis (preprocessing, processing, and classification). This GUI will abstract the complexities of code-based toolboxes like FieldTrip, thereby lowering the technical barrier to entry and welcoming new users to the field of neuroscience and EEG analysis.
- **Research Goal (Practical Outcomes):** To validate the utility and efficacy of the developed GUI and its integrated deep learning classification pipeline by generating meaningful, publishable findings. This includes providing example EEG data outcomes, specifically robust analysis results and accurate classification metrics for clinical datasets concerning Parkinson's Disease.

The ultimate goal is to deliver a practical product that enhances the efficiency of the EEG analysis pipeline and produces high-quality, reproducible research outcomes.

I.4 Data Description

I.4.1 Parkinson's

Chapter II

Methodology

II.1 System Architecture & Workflow

The NeuroPAC platform is structured into three primary modules that collectively facilitate a comprehensive EEG data analysis pipeline: Preprocessing, Analysis, and Classification. Each module is designed to handle specific stages of the workflow, ensuring a systematic approach to EEG data handling and interpretation.

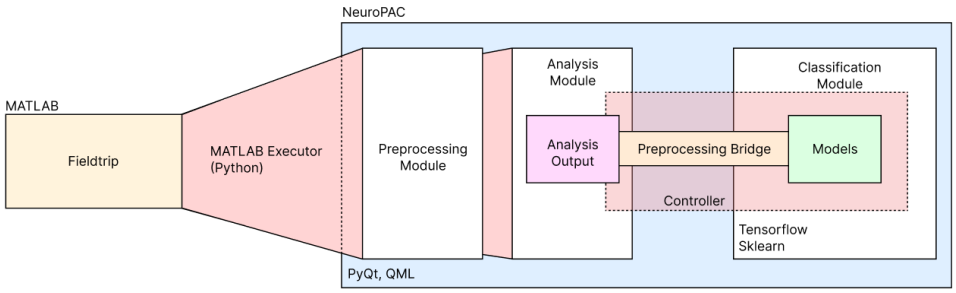


Figure II.1: System Architecture of NeuroPAC platform illustrating the three main modules: Preprocessing, Analysis, and Classification.

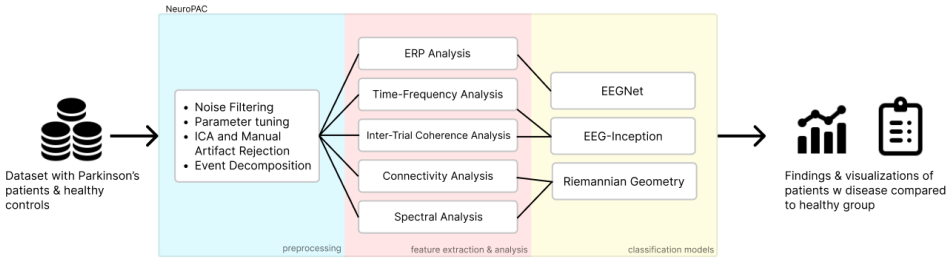


Figure II.2: Workflow of NeuroPAC platform illustrating the sequence of operations across the three main modules: Preprocessing, Analysis, and Classification.

II.2 Preprocessing Module

II.2.1 Preprocessing

Preprocessing is the very first step of the EEG analysis pipeline. It includes adjusting the raw EEG data to make it suitable for further analysis. To analyze a dataset, we must first initialize a cfg data structure from the FieldTrip toolbox, which serves as our data container and provides numerous configurable properties. The preprocessing module of NeuroPAC streamlines this process by allowing users to set cfg structure properties directly through the GUI, automating the workflow without requiring manual code configuration. The preprocessing steps currently implemented in NeuroPAC as base preprocessing components are as follows:

- `cfg.trialfun = 'ft_trialfun_general'` : Defines trials based on specified events and time windows.
- `cfg.trialdef.eventtype = 'stimulus'`: Specifies the type of event to use for trial definition.
- `cfg.trialdef.eventvalue = {'S200', 'S201', 'S202'}` : Sets the specific event value to identify trials (conditions).
- `cfg.channel = 'all'` : Selects all available EEG channels for analysis.
- `cfg.demean = 'yes'` : Removes the mean value from each channel.
- `cfg.baselinewindow = [-0.2 0]` : Defines the time window for baseline correction.
- `cfg.dftfilter = 'yes'` : Enables the use of a notch filter to remove line noise.
- `cfg.dftfreq = [50 60]` : Specifies the frequencies to be removed by the notch filter (e.g., 50 Hz and its harmonics).

These steps can be deleted, modified, or new ones can be added from the top menu as needed. These are individual lines that get converted from simple MATLAB script lines to specific QML template items by the parameter parser, you can read further about the parameter

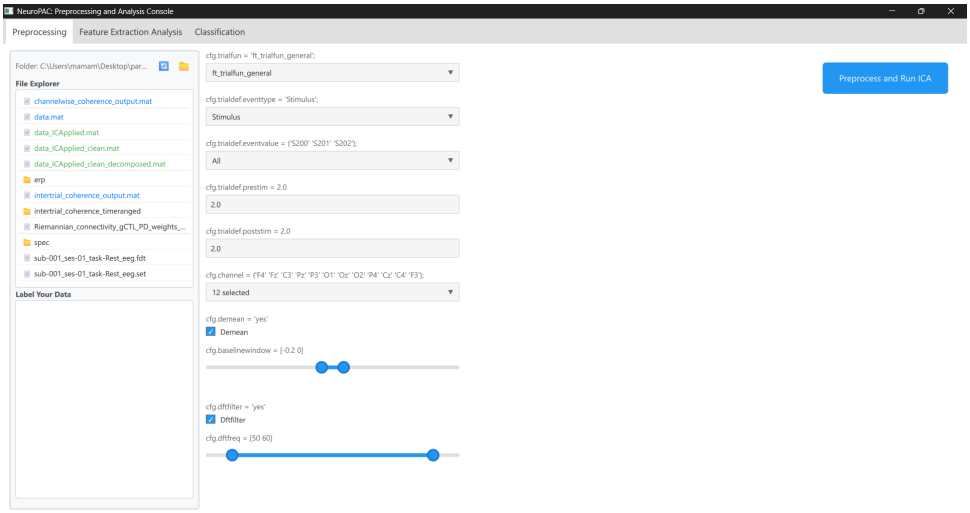


Figure II.3: Preprocessing module interface showing configurable parameters and MATLAB script generation.

parser and template items themselves on sections 3.3.2 and 3.3.3. Before moving on to running preprocessing, the user has to make sure that:

- Their fieldtrip path is correctly set from the top menu,
- and the file browser is pointing to the correct directory where the raw EEG data is located.

After these configurations are set, the user can run the `Preprocess and Run ICA` button, which executes a Fieldtrip function called `ft_preprocessing` that applies the defined preprocessing steps to the raw EEG data. This operation will require to run on MATLAB, therefore the MATLAB will open to ensure a smooth execution and will provide any warnings from its console that the user might need to be aware of. After the preprocessing is done, the preprocessed data will be saved in a file named `data.mat` in the same directory as the raw data. After preprocessing is done, the button will fulfill its second purpose, which is to run ICA on the preprocessed data. ICA is a computational dimension reduction method used to separate a multivariate signal into additive, independent components. In the context of EEG data, ICA is commonly employed to identify and remove artifacts (unwanted signals) such as eye blinks, muscle activity, and other noise sources that can contaminate the EEG recordings. In our case ICA is chosen as `fast_ica` algorithm provided by Fieldtrip, and in this study's case only the eye artifacts were considered and removed. After running ICA, the resulting components will be saved in a file named `data_ICApplied.mat` in the same directory as the raw data for further

inspection and artifact removal.

II.2.2 Independent Component Analysis (ICA) & Artifact Removal

Target Control Group vs Parkinson's Disease

Table II.1: EEGNet ERP classification performance (target vs Parkinson's Disease)

tSplit	Accuracy (%)	Loss
Training	63.33	0.6884
Validation	60.00	0.6972
Test	50.00	0.6921

Standard Control Group vs Parkinson's Disease

Table II.2: EEGNet ERP classification performance (standard vs Parkinson's Disease)

tSplit	Accuracy (%)	Loss
Training	86.67	0.5161
Validation	80.00	0.5632
Test	70.00	0.7055

Novelty Control Group vs Parkinson's Disease

The user, if ready to proceed, can then move on to the artifact removal step by clicking the `data_ICApplied.mat` file name directly from the file browser, which will open a matlab window displaying a slightly modified version of `ft_databrowser` which now contains a "Reject ICA" button on the right of the browser window. If any user wants to replicate the same application experience, they would have to run the syncing script, which is given as an option in the top menu. This will help to modify their own "`ft_databrowser`" script by adding the "Reject IC" button. The independent components obtained from ICA will be shown and iterated for every subject. The user can then visually inspect these components and select the ones corresponding to artifacts (e.g., eye blinks) for removal with the "Reject IC" button. Once the user selects the components to be removed and confirms their choice, the array with the

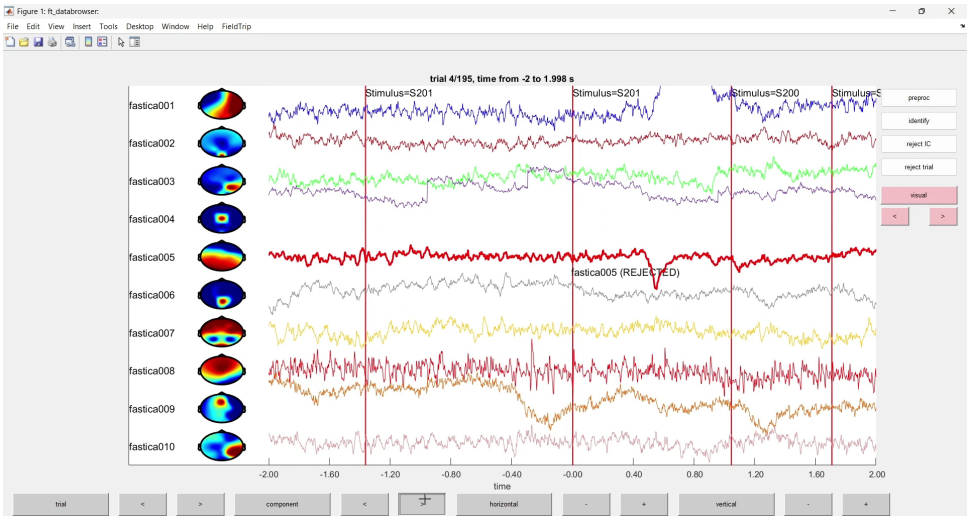


Figure II.4: `ft_databrowser` interface with the externally added Reject IC button for artifact removal, while rejecting IC 5 on a subject.

rejected IC's will be saved and then be iterated through the whole data, removing each rejected IC of the corresponding subject. The system will reconstruct the EEG data without the selected artifact components.

II.2.3 Event/Condition Decomposition

After the IC rejection, a quick event (trial type) decomposition is made to ready the final data for analyses. The decomposition's job is to decompose the trials by their corresponding numbers 200, 201, 202 to target, standard and novelty respectively. The outcome of this final process results in a subject number $\times$ 1 struct with three fields (three being the trial conditions), and then the cleaned and decomposed data is saved in a file named `data_ICApplied_clean_decomposed.mat` in the same currently open file browser directory. This cleaned and decomposed data is now ready for feature extraction and analysis in the Analysis Module.

II.3 Analysis Module

The analysis module of NeuroPAC provides a suite of commonly used EEG analysis techniques that can be easily configured and executed through the GUI. The module allows users to select from various analysis methods, set relevant parameters, and visualize the results. The available analyses include:

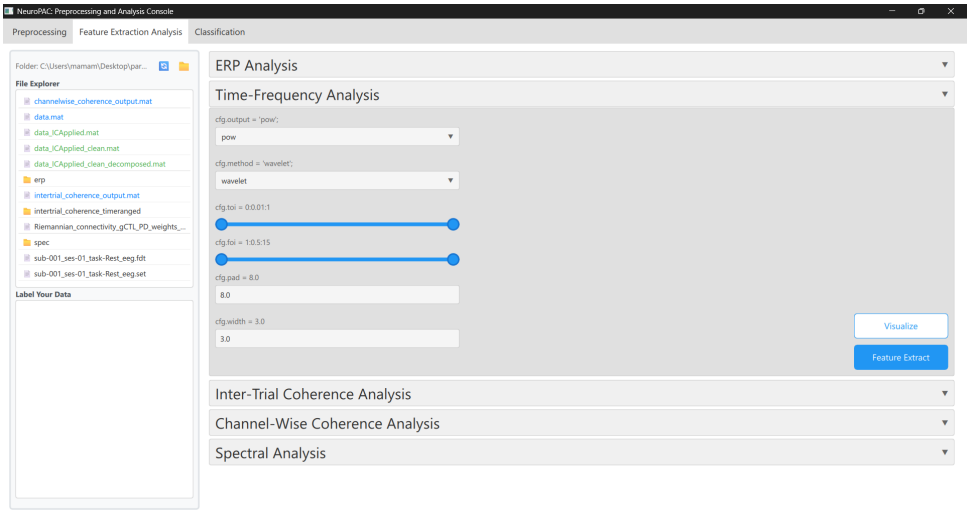


Figure II.5: Analysis module interface showing configurable analysis options and MATLAB script generation.

- Event-Related Potential (ERP) Analysis,
- Time-Frequency Analysis,
- Spectral Analysis,
- Channel-Wise Connectivity Analysis,
- Inter-Trial Coherence Analysis.

These analyses are commonly used in EEG research and provide a solid foundation for understanding various aspects of EEG data. Each analysis serves a specific purpose and can be further customized or expanded, or can be further feature-extracted before the classification step based on the user's research needs.

II.4 Common Procedures During Analyses

All analysis modules in NeuroPAC follow a standardized procedural framework to ensure consistency, reliability, and efficient resource management across different analysis types. The key shared components are:

II.4.1 File Management and Error Handling

Analysis functions search for the standard preprocessing output file `data_ICApplied_clean_decomposed.ma` accepting either direct file paths or folder paths. The functions intelligently extract parent directories if needed and verify the existence of the expected `clean_data_decomposed` variable, providing clear error messages if data integrity issues arise. All file operations are wrapped in try-catch blocks for robust error handling.

II.4.2 FieldTrip Initialization

Each analysis script automatically detects and initializes FieldTrip if not already available on the MATLAB path. The detection routine first checks configuration files from the preprocessing module for previously specified paths, then searches common installation directories. If FieldTrip is located, the script adds it to the path and calls `ft_defaults`. If initialization fails, the script provides clear error messages with manual setup instructions.

II.4.3 Batch Processing for Memory Efficiency

To handle large datasets without exhausting system memory, all analysis scripts process subjects in batches of ten. This approach allows analyses to complete successfully on systems with limited resources while maintaining computational efficiency. Console output marks the beginning and completion of each batch, and subjects within each batch are processed sequentially across all three experimental conditions (target, standard, novelty).

II.4.4 Baseline Correction

Following primary analysis computations, all applicable analyses apply baseline correction using FieldTrip's dedicated functions. The baseline correction normalizes measurements relative to pre-stimulus activity, ensuring observed responses reflect stimulus-induced changes rather than pre-existing offsets. Baseline parameters vary by analysis type but follow a consistent approach: defining a pre-stimulus window, computing mean values, and subtracting these from

post-stimulus periods. Note that some analyses (spectral and connectivity-based) omit baseline correction when the measure itself provides normalized outputs.

II.4.5 Result Storage and Output

All analysis scripts save results in well-organized data structures arranged by subject and condition, including essential metadata (time vectors, frequency vectors, channel labels). Output files use descriptive naming conventions clearly indicating the analysis type and are saved alongside the source data. Throughout processing, scripts maintain clear console communication confirming completion stages and providing final output file locations.

II.4.6 ERP Analysis

Event-Related Potential (ERP) analysis examines brain responses time-locked to experimental stimuli. In NeuroPAC, the `timelock_func.m` script implements ERP analysis using FieldTrip's timelock analysis functions following the common procedural framework.

Configuration and Computation

The analysis configures a time window from 0 to 1 second post-stimulus using the parameter `cfg.latency`. FieldTrip's `ft_timelockanalysis` function averages individual trials within each condition to produce the characteristic ERP waveform, enhancing signal-to-noise ratio by reinforcing time-locked neural activity. The script processes target, standard, and novelty conditions conditionally, skipping empty data fields.

Baseline Correction and Output

Baseline correction uses a pre-stimulus window from -0.2 to 0 seconds, representing the 200 milliseconds immediately preceding stimulus onset. This short baseline period captures the immediate pre-stimulus state while avoiding contamination from earlier events. The correction employs absolute subtraction via FieldTrip's `ft_timelockbaseline` function.

Results are reformatted for compatibility with visualization modules, creating a subjects-by-conditions matrix containing time vectors, averaged voltage values, channel labels, variance

measures, and degrees of freedom. The script saves two complementary structures: `ERP_data` (detailed timelock output) and `erp_records` (reformatted for visualization) in `erp_output.mat`.

II.4.7 Time-Frequency Analysis

Time-frequency analysis reveals how spectral content evolves over time, preserving both temporal and frequency dimensions simultaneously. This complements ERP analysis (time-domain) and spectral analysis (frequency-domain) by capturing dynamic changes in neural oscillations. In NeuroPAC, the `timefreqanalysis.m` script implements wavelet-based decomposition following the common procedural framework.

Data Structure Normalization

The script inspects and normalizes input data to handle varying organizational formats from different preprocessing workflows. Regardless of whether data uses field names like `target`, `target_data`, or matrix organization, the normalization produces a consistent structure where each subject contains three condition fields (target, standard, novelty), enabling uniform processing.

Wavelet Configuration and Computation

The analysis employs wavelet decomposition with the following configuration: frequency range from 1 to 15 Hz (half-hertz increments, covering delta/theta/alpha bands), time points from 0 to 1 second post-stimulus (10-millisecond steps), wavelet width of 3 cycles, and 8-second padding to prevent edge artifacts. FieldTrip's `ft_freqanalysis` function computes power spectra at each time-frequency point using complex Morlet wavelets, capturing event-related synchronization and desynchronization patterns.

Baseline Correction and Output

Baseline correction uses a pre-stimulus window from -1 to -0.5 seconds, providing a longer baseline than ERP analysis for more stable power estimates across frequency bands. The correction applies absolute subtraction via FieldTrip's `ft_freqbaseline` function. Results are

packaged into `timefreq_data` (organized by subject and condition, including power values, time vectors, frequency vectors, and channel labels) and saved to `timefreq_output.mat`.

II.4.8 Spectral Analysis

Spectral analysis examines frequency content averaged across entire trial epochs, complementing time-frequency analysis which preserves temporal dynamics. This approach characterizes baseline oscillatory properties and frequency-specific changes sustained throughout experimental conditions. In NeuroPAC, the `spectralanalysis.m` script implements Fourier-based decomposition.

Data Structure Normalization and Configuration

The script implements the same data structure inspection and normalization phase used in time-frequency analysis, handling various organizational formats and producing a consistent structure where each subject contains three condition fields (target, standard, novelty). The configuration specifies the multitaper fast Fourier transform method (`mtmfft`) with Fourier coefficients as output (preserving phase information for downstream connectivity analyses), Hanning taper windowing, and frequency range from 1 to 15 Hz (half-hertz increments, covering delta/theta/alpha bands).

Computation and Output

FieldTrip's `ft_freqanalysis` function applies the multitaper Fourier transform to all trials for each subject-condition combination with valid data, computing Fourier coefficients for each frequency bin, channel, and trial. Unlike ERP and time-frequency analyses, spectral analysis omits baseline correction because coherence inherently measures relative phase consistency and Fourier coefficients represent spectral composition without requiring baseline normalization. Results are packaged into `spectral_data` (organized by subject and condition, containing Fourier coefficients, frequency vectors, channel labels, and metadata) and saved to `spectral_output.mat`.

II.4.9 Channel-Wise Connectivity Analysis

Channel-wise connectivity analysis investigates functional relationships between different brain regions by quantifying how neural signals from different electrode locations interact with each other. Unlike signal-level analyses that examine individual channels in isolation, connectivity analysis characterizes the synchronization of neural oscillations across the scalp. In NeuroPAC, the `channelwise.m` script computes coherence measures between all pairs of EEG channels.

Data Normalization and Three-Stage Pipeline

The script implements data structure normalization to handle various organizational formats from preprocessing workflows. The analysis requires a three-stage computational pipeline: (1) frequency decomposition using wavelet analysis with Fourier coefficient output (preserving phase information), (2) temporal selection specifying a latency window from 0 to 1 second to exclude edge artifacts, and (3) coherence computation quantifying phase relationship consistency between channel pairs. The wavelet configuration mirrors time-frequency analysis (frequency range 1-15 Hz in half-hertz increments, 10-millisecond time steps, 3-cycle wavelet width, 8-second padding).

Coherence Computation

FieldTrip's `ft_freqanalysis` produces time-frequency representations with Fourier coefficients, then `ft_selectdata` extracts the specified latency window, and `ft_connectivityanalysis` computes coherence values between all channel pairs. Coherence (range 0-1) measures the consistency of phase relationships across trials; values near zero indicate weak functional coupling while values near one indicate strong phase synchronization. Unlike correlation-based measures, coherence examines frequency-resolved phase relationships, avoiding volume conduction artifacts.

Output

Coherence analysis omits baseline correction because coherence inherently measures relative phase consistency without reference to absolute signal amplitudes. Results are packaged

into `coherence_data` (organized by subject and condition, containing coherence matrices for all channel pairs across frequencies, channel labels, and frequency vectors) and saved to `channelwise_coherence_output.mat`, enabling examination of both local and long-range connectivity patterns.

II.4.10 Inter-Trial Coherence Analysis

Inter-trial coherence analysis quantifies the extent to which neural oscillations are phase-locked to experimental events, examining phase consistency across trials at the same channel location rather than across different channels. This reveals whether oscillatory activity timing relative to stimulus onset is consistent across repeated presentations. High inter-trial coherence indicates stimulus-driven entrainment of neural oscillations, while low values indicate variable phase relationships. In NeuroPAC, the `intertrialcoherenceanalysis.m` script implements phase-locking quantification.

Data Normalization and Configuration

The script implements the same data structure inspection and normalization phase used in other frequency-domain analyses, handling various organizational formats with adaptive field-selection logic. The configuration parallels channel-wise connectivity analysis: wavelet-based frequency decomposition with Fourier coefficient output (preserving phase information), frequency range from 1 to 15 Hz (half-hertz increments), time points from 0 to 1 second (10-millisecond steps), wavelet width of 3 cycles, and 8-second padding. Fourier coefficients are essential because inter-trial coherence fundamentally depends on phase information.

Two-Stage Computational Pipeline

The analysis executes a two-stage pipeline for each subject-condition combination. First, FieldTrip's `ft_freqanalysis` produces a time-frequency representation with Fourier coefficients for each channel, frequency, time point, and trial. Second, a dedicated `compute_itc` function examines Fourier coefficients for all trials at each channel, frequency, and time point, computing the degree to which phase angles are aligned across the trial ensemble. The resulting

inter-trial coherence values (range 0-1) indicate phase-locking strength; zero represents completely random phase relationships while one indicates perfect phase alignment across all trials.

Distinction from Channel-Wise Coherence and Output

Inter-trial coherence and channel-wise coherence measure fundamentally different phenomena. Channel-wise coherence quantifies phase synchronization between two spatial locations across trials; inter-trial coherence quantifies phase synchronization across trials at a single location. Channel-wise coherence reveals spatial organization of brain networks, while inter-trial coherence reveals temporal precision of stimulus-evoked responses. Like other frequency-domain analyses, inter-trial coherence omits baseline correction because the measure inherently quantifies relative phase consistency.

Results are packaged into two complementary structures: `itc_data` (standard structure array organized by subject and condition) and `itc_records` (cell array format for flexible downstream operations). Both are saved to `intertrial_coherence_output.mat`, providing organizational flexibility for different analytical needs.

II.5 Classification Module

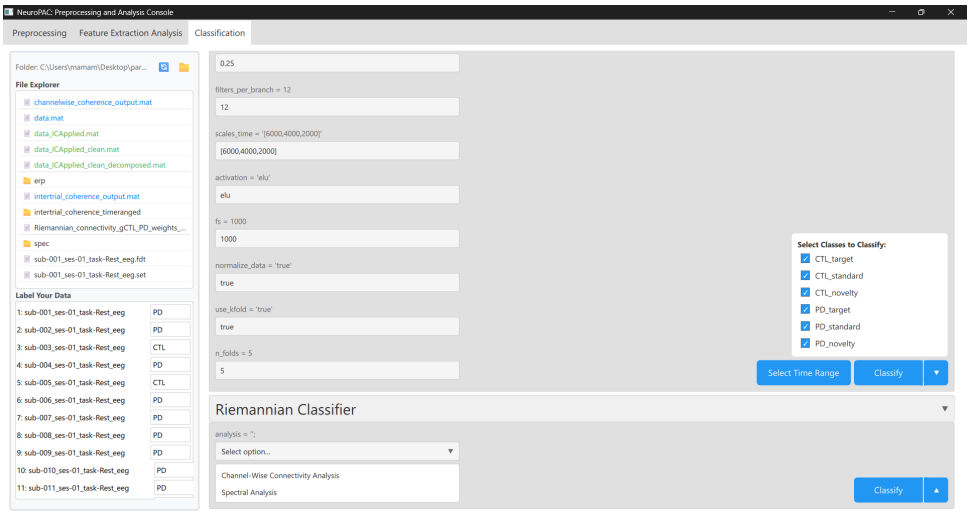


Figure II.6: Classification module interface showing configurable hyperparameters and classification options.

II.5.1 EEGNet Classifier

EEGNet is a compact deep learning architecture designed specifically for EEG signal classification. Originally introduced by Lawhern et al. in 2018, the model addresses the unique challenges of EEG data by incorporating separate temporal and spatial filtering stages that mirror traditional signal processing approaches while remaining parameter-efficient for limited training data scenarios.

Configuration Parameters

Key parameters control the model architecture:

- `nb_classes`: Number of classification categories
- `Chans`, `Samples`: Number of EEG channels and time points per trial
- `F1`, `D`, `F2`: Filter counts for temporal, spatial, and separable convolution stages
- `kernLength`: Temporal extent of initial filters (recommended: half the sampling rate)
- `dropoutRate`, `norm_rate`: Regularization parameters controlling overfitting
- `dropoutType`: Choice between Dropout and SpatialDropout2D based on data type

Architecture Overview

The input tensor is organized as channels by time points by a singleton feature dimension, treating channels analogously to image height and time analogously to width. The architecture progresses through two primary convolutional blocks followed by a classification layer.

First Block: Temporal and Spatial Filtering

The first block learns both temporal and spatial patterns. A temporal convolution with `F1` filters spans one channel by `kernLength` time points, detecting temporal patterns across all channels simultaneously. This is followed by batch normalization and a depthwise convolution that applies spatial filters of all channels by one time point, creating `F1 * D` feature maps. The `depthwise_constraint` regularization limits filter norms to one, preventing individual filters

from dominating. An exponential linear unit (ELU) activation introduces nonlinearity. Average pooling reduces temporal resolution by a factor of four, and dropout prevents overfitting.

Second Block: Feature Combination

The second block uses separable convolutions to optimally combine spatial features across the temporal dimension. Depthwise filters span one by sixteen time points, followed by a pointwise convolution with `F2` filters that learns weighted combinations of features. A more aggressive average pooling (kernel size one by eight) further reduces temporal resolution, creating abstract representations of longer-range dependencies.

Classification Layer

A flattening operation concatenates all feature dimensions into a one-dimensional vector. A single fully connected dense layer maps this to the output space with softmax activation. The `kernel_constraint` parameter regularizes the classification weights.

Design Rationale

EEGNet's effectiveness derives from its careful architectural decisions. By separating temporal and spatial filtering, the model learns interpretable features aligned with neuroscientific understanding. Parameter efficiency enables effective training with limited data. The use of depthwise separable convolutions reduces computational cost while maintaining representational capacity. The constraint-based regularization (norm constraints on depthwise and classification layers) prevents overfitting by encouraging diverse, distributed feature representations rather than reliance on a few dominant filters.

II.5.2 EEG-Inception Classifier

EEG-Inception employs parallel multi-scale processing that simultaneously examines input data at different temporal resolutions, contrasting with EEGNet's sequential temporal-then-spatial approach. This architecture recognizes that neural signals contain information at multiple time scales, from fast transient responses to slower oscillatory patterns. NeuroPAC provides

three variants: a base model for raw time series, a time-frequency variant for spectrograms, and an inter-trial coherence variant for phase consistency patterns.

Base Model Configuration

Key parameters control the architecture:

- `input_time`, `fs`: Epoch duration (ms) and sampling frequency; together determine temporal samples
- `ncha`: Number of EEG channels
- `filters_per_branch`: Filters per parallel processing branch
- `scales_time`: List of temporal scales in milliseconds (default: 500, 250, 125 ms)
- `dropout_rate`, `activation`, `n_classes`, `learning_rate`: Standard regularization and training parameters

Architecture Overview

The input tensor is organized as temporal samples by channels by a singleton feature dimension. The architecture progresses through multiple Inception blocks that progressively compress and refine multi-scale features.

First Inception Block

Creates parallel processing pathways for each temporal scale specified. Each pathway applies: (1) temporal convolution spanning the full scale by one channel with He normal initialization, (2) batch normalization, (3) ELU activation, (4) depthwise convolution with depth multiplier of two and max-norm constraint of one, (5) batch normalization and dropout. All pathways are concatenated along the feature dimension, producing a unified multi-scale representation. Adaptive average pooling further reduces temporal resolution while preventing degenerate dimensions.

Second Inception Block

Operates on pooled features from the first block using reduced kernel sizes (one-quarter of original scales). Excludes depthwise spatial convolutions, focusing instead on combining and abstracting spatially-filtered features from the first block. Concatenated outputs undergo adaptive pooling for further downsampling.

Classification Stages

A sequential refinement pathway progressively reduces feature dimensionality through two convolutional stages with adaptive kernel sizes and filter counts that create a bottleneck, forcing the network to learn the most discriminative features. Flattening converts feature maps to a one-dimensional vector, and a dense layer with softmax produces class probabilities. Training uses Adam optimizer with specified learning rate and binary cross-entropy loss.

Time-Frequency Variant

Operates on spectrogram data (frequency bins $\times$ time points $\times$ channels) from wavelet decomposition. Parameters `n_freqs`, `n_times`, `n_channels` specify dimensions (defaults: 91, 29, 12). Multi-scale kernel sizes are defined in frequency (`scales_freq`: defaults 3, 5, 9) and time (`scales_time`: defaults 3, 5, 7) dimensions.

More aggressive regularization includes spatial dropout (0.15) after input and channel expansion via one-by-one convolutions. Inception blocks employ separable convolutions (decomposed into depthwise and pointwise) with 12 filters per branch. Classification consists of dense layers (128, 64 units) with batch normalization and dropout (0.4, 0.3).

A critical innovation uses focal loss instead of standard cross-entropy, emphasizing hard-to-classify examples. The modulating factor downweights well-classified examples, with focusing parameter $\gamma = 4.0$ creating strong emphasis on difficult cases. This proves valuable in medical diagnostic applications with class imbalance.

Inter-Trial Coherence Variant

Mirrors the time-frequency architecture but adds class-specific weighting via the `class_weights` parameter. The `calculate_class_weights` function automatically computes weights based

on class frequency: *balanced* strategy applies inverse frequency weighting, while *aggressive* strategy amplifies minority class weights by raising ratios to power 1.5. Weights are normalized so the mean equals one. Focal loss integrates these weights, allowing examples from high-weight classes to influence training more strongly, handling severe class imbalance and varying diagnostic importance.

Architectural Variants

Both time-frequency and inter-trial coherence variants provide simplified alternatives:

- **V2 variants**: Implement residual connections (ResNet-inspired) with shortcut pathways facilitating gradient flow through deeper networks
- **Lightweight variants**: Shallower architectures with fewer layers and smaller filters for small datasets
- **Simple variants**: Minimal architectures with single convolutional blocks or direct classification, suitable when training data is extremely limited

Model complexity is matched to data availability, with simpler models preferred when data is scarce.

Design Philosophy

EEG-Inception's parallel multi-scale processing enables the network to learn which temporal scales matter most for a given task, rather than making predetermined assumptions. The availability of variants for different input formats (raw time series, spectrograms, phase consistency) reflects that EEG can be represented multiple ways, and optimal architectures must respect each format's statistical properties. Advanced training techniques (focal loss, class weighting) address practical challenges in medical diagnosis. The flexibility to match model complexity to data characteristics creates a framework capable of handling diverse EEG analysis scenarios while remaining grounded in neuroscientific principles.

II.5.3 Riemannian Classifier

The Riemannian classifier employs differential geometry and manifold theory to classify EEG data, contrasting with deep learning approaches. It recognizes that covariance matrices derived from multi-channel EEG do not exist in Euclidean space but on a curved Riemannian manifold (symmetric positive definite matrices). By employing geometric operations that preserve manifold structure, this approach often achieves superior classification performance. NeuroPAC provides two variants: a base implementation for covariance matrices and a connectivity-focused implementation for functional connectivity measures.

Theoretical Foundation

Covariance matrices satisfy two fundamental properties: symmetry and positive definiteness (all eigenvalues strictly positive). These constraints define a curved manifold rather than flat Euclidean space. Euclidean operations fail to respect this geometry; for example, arithmetic means of covariance matrices in Euclidean fashion may not be optimal on the manifold, and Euclidean distances misrepresent geometric relationships. The Riemannian framework provides mathematical tools to work directly on the manifold, defining appropriate metrics, geodesics, and means that respect intrinsic geometry.

Base Pipeline Configuration

Key parameters control the pipeline:

- `estimator`: Covariance estimation method (sample covariance, Ledoit-Wolf shrinkage, oracle approximating shrinkage, minimum covariance determinant)
- `metric`: Riemannian metric for geometric operations (affine-invariant `riemann`, logarithmic Euclidean `logeuclid`, standard `euclid`)
- `use_mdm`: Enable Minimum Distance to Mean classifier
- `use_deep`: Enable deep learning classification via multilayer perceptron
- `kernel`, `C`: SVM kernel type and regularization when using SVM classification

- Deep learning parameters: `hidden_dims`, `dropout`, `lr`, `max_epochs`, `batch_size`, `weight_decay`, `patience`

Pipeline Stages

Covariance Estimation

The Covariances transformer accepts raw EEG data (trials $\times$ channels $\times$ time points) and produces covariance matrices (trials $\times$ channels $\times$ channels), computing second-order statistics characterizing spatial relationships. Shrinkage-based estimators (Ledoit-Wolf, oracle approximating shrinkage) provide stable estimates when channels approach or exceed temporal samples, preventing ill-conditioned matrices. Each covariance matrix represents a trial as a point on the manifold, encoding spatial correlation structure while discarding temporal dynamics.

Tangent Space Projection

For classical or deep learning classification, the TangentSpace transformer linearizes curved manifold geometry into flat Euclidean space suitable for conventional algorithms. A reference point (typically the Riemannian mean of training covariance matrices) serves as the tangent space origin. The Riemannian metric computes this mean via iterative optimization minimizing squared geodesic distances; log-Euclidean metric uses closed-form solutions via matrix logarithms. Each covariance matrix maps to the tangent space via logarithmic mapping, computing the tangent vector along the geodesic connecting reference and target. Resulting tangent vectors have dimensionality $n_{channels}(n_{channels}+1)/2$, creating Euclidean feature vectors that preserve local geometric relationships while enabling standard machine learning operations.

Classification Methods

Minimum Distance to Mean (MDM): Computes a Riemannian mean for each class, classifying new trials by geodesic distance to class means. This pure geometric approach respects manifold structure throughout, avoiding linearization approximations. Effective when classes exhibit distinct covariance patterns forming separated clusters on the manifold and training samples per class are sufficient for stable mean estimation. Higher computational cost due to eigenvalue decompositions and matrix logarithms compared to tangent space methods.

Deep Learning: The TFMLPClassifier receives tangent space features after standardization (zero mean, unit variance). Network architecture includes input dropout, hidden layers with dense transformations, ReLU activation, batch normalization, and dropout, with optional L2 weight decay. Training uses Adam optimizer and sparse categorical cross-entropy loss. Early stopping with validation monitoring prevents overfitting. This hybrid approach combines Riemannian feature extraction with neural network classification.

Support Vector Machine: Operates on standardized tangent space features, constructing maximum-margin hyperplanes. The `kernel` parameter determines transformations: linear kernel for linearly separable classes, radial basis function for complex nonlinear boundaries, polynomial kernels for polynomial decision surfaces. The `C` parameter controls margin-error trade-off. Class weighting handles imbalanced datasets with inverse frequency scaling. Probability estimation fits logistic regression on SVM decision values.

Connectivity-Focused Implementation

Functional connectivity matrices (coherence, phase locking value, correlation) may violate positive definiteness assumptions due to estimation noise or singular conditions, causing standard Riemannian methods to fail. The connectivity pipeline introduces regularization ensuring all matrices satisfy geometric requirements while preserving discriminative information.

The ConnectivityRegularizer transformer applies multi-stage regularization: (1) symmetry correction by averaging with transpose, (2) diagonal regularization adding small positive values to shift eigenvalues upward, (3) shrinkage toward identity providing numerical stability, (4) optional eigenvalue clipping explicitly enforcing positive definiteness. The `preserve_scale` option maintains matrix trace, preventing systematic magnitude distortion.

The SoftTangentSpace transformer extends standard tangent space projection with error handling and fallback mechanisms. Pre-regularization strengthens mean computation stability. Exception handling catches failures and falls back to log-Euclidean or Euclidean operations when full Riemannian approach encounters numerical difficulties, ensuring pipeline completion while noting fallback activation for data quality investigation.

Alternative vectorized mode bypasses Riemannian geometry, extracting the upper triangle

of connectivity matrices as feature vectors. The ConnectivityVectorizer applies optional logarithmic transformation improving distributions when values span multiple orders of magnitude. This approach sacrifices geometric principles for simplicity and efficiency, proving effective when manifold geometry contributes minimally or matrices are severely corrupted. Deep learning classification operates on vectorized features with more aggressive regularization (dropout 0.75, smaller hidden layers, reduced batch size 8) appropriate for small connectivity datasets.

Advantages and Limitations

Advantages: Interpretability through covariance/connectivity patterns directly relating to functional brain relationships; parameter efficiency enabling effective classification with smaller training sets since covariance estimation dramatically reduces dimensionality; mathematical elegance through principled geometric handling.

Limitations: Loss of temporal information through static covariance matrices; stationarity assumptions implicit in covariance estimates; sensitivity to estimation quality, particularly problematic with short trials or high channel counts.

Flexibility to choose pure geometric methods (MDM, best with limited data) or hybrid approaches (combining Riemannian extraction with deep learning, advantageous with sufficient data) enables adaptation to dataset characteristics. The connectivity variant with regularization extends applicability to practical scenarios where strict positive definiteness is violated.

II.6 Other Application Components

II.6.1 Template Items

Range Slider Template



Figure II.7: Range slider component for selecting numeric intervals.

The range slider QML component displays a pair of draggable handles to select a numeric interval, keeps text fields in sync for min/max/current values, and snaps changes to a configurable step with sensible rounding. It supports an edit/add mode that lets users rename the bound MATLAB property, preview the exact assignment text (including special handling for trial windows), and persist or delete the property via hooks to the global `matlabExecutor`. Input validation prevents inverted ranges, enforces bounds, and surfaces warnings inline; helper functions also update related MATLAB settings when specialized slider IDs (baseline, dftfreq, prestim/poststim, ERP) are used. Visual affordances include background highlighting of the active span, focus-driven z-ordering, and icon buttons for saving or removing the parameter while respecting read-only vs. edit states.

Step Range Slider Template

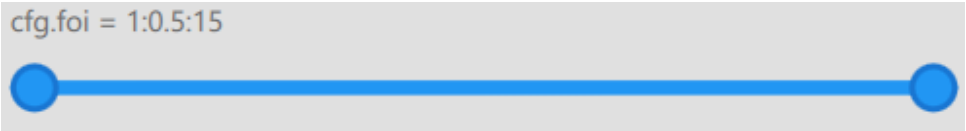


Figure II.8: Step range slider component for selecting numeric intervals with defined steps.

This stepped range slider renders a double-handled control for values defined in MATLAB's start:step:end syntax, keeps paired text inputs synchronized for bounds, step size, and handle positions, and snaps all changes to the current step with precision-aware rounding. An add/edit mode lets users rename the underlying `cfg` property, preview the exact colon-form assignment, and persist or delete it via `matlabExecutor` hooks; validation guards against reversed bounds, zero/negative steps, and out-of-range entries with inline warnings. When slider IDs match known preprocessing roles (baseline, dftfreq, prestim/poststim, ERP) it calls specialized executor updaters; otherwise it saves dynamic parameters through the step-range MATLAB writer, ensuring UI edits stay in lockstep with script values.

Dropdown Menu Template

The dropdown component renders both single- and multi-select pickers, keeping a MATLAB-bound property in view with editable labels and add/delete actions. It hides the stock controls

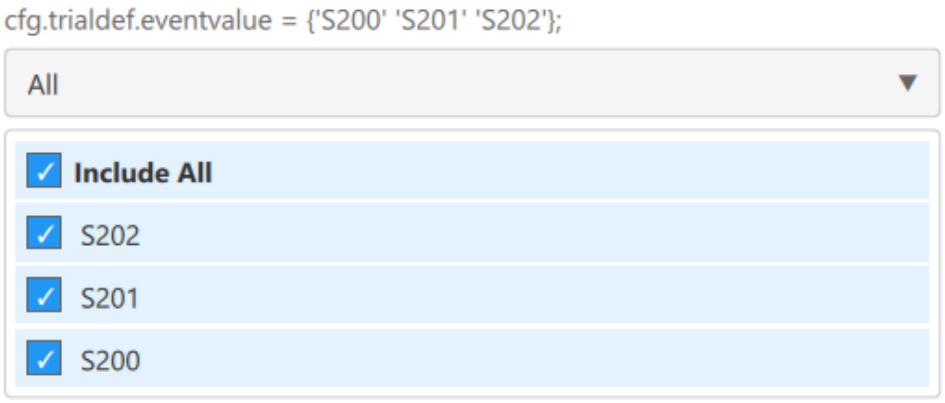


Figure II.9: Dropdown menu component showing multi-select options with checkboxes.

and replaces them with custom popups, optional checkboxes, and an "add custom" flow that persists new options. For multi-select it summarizes selections ("All", count, or none) while exposing the full quoted list in the assignment preview; single-select mirrors the current choice. An edit/add mode lets users rename the `cfg` property, save via `matlabExecutor`, or remove the assignment entirely. Z-index management, focus handling, and inline validation keep the control usable in dense layouts.

Input Box Template

The input box presents a single-line field bound to a MATLAB property, with optional numeric mode, placeholder text, and a compact display of the exact assignment (quoted for strings, raw for numbers). Edit/add mode lets users rename the `cfg` key, preview the assignment, and save or delete it through `matlabExecutor` hooks; focus and z-ordering keep the control usable in dense layouts. Text changes emit live signals so downstream logic can react immediately, while a save icon commits the value and a trash icon removes the property when exposed in edit contexts.

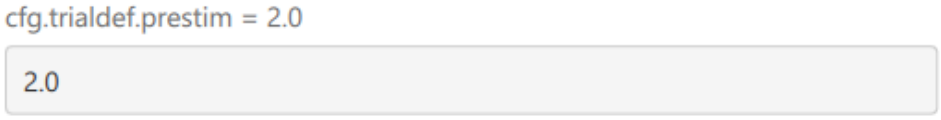


Figure II.10: Input box component for string/numeric MATLAB property binding.

Analysis Module Template

The Analysis Module Template is a reusable QML component providing a consistent interface for configuring and executing MATLAB-based analyses. It encapsulates complex functionality into a self-contained, configurable unit supporting any analysis type while maintaining visual and behavioral consistency. The template handles the complete lifecycle from parameter configuration through execution and visualization.

Core Architecture

The component implements a collapsible panel design conserving screen space. Initially collapsed, showing only a header bar with the analysis name, it expands on click to reveal parameter controls and action buttons. This accordion-style pattern allows multiple analysis modules on one page without overwhelming users, letting them focus on one analysis at a time while remaining aware of others. Height dynamically adjusts based on expansion state.

Parameter Management and Dynamic Rendering

When instantiated, the template leverages the MATLAB parameter parser through the `matlabExecutor` object, which analyzes the corresponding MATLAB script and extracts all configurable parameters with their types, defaults, and ranges. This information returns as JSON, deserialized into the `dynamicParameters` property serving as the data model for interface generation.

A Repeater iterates over all parameters, delegating rendering to a DynamicParameterLoader that examines metadata and selects appropriate control types: range sliders for numeric ranges, step range sliders for stepped sequences, dropdowns for enumerations, input boxes for strings/numbers, and checkboxes for booleans. Each control receives configuration directly from parameter metadata, automatically matching interface constraints to MATLAB script requirements without manual synchronization.

Workflow Controls

Two action buttons become visible when expanded:

Feature Extract Button: Primary action triggering analysis computation via a `buttonClicked` signal. Uses solid blue styling emphasizing its role. Always visible when expanded, allowing

users to execute or re-execute analyses with current parameters.

Visualize Button: Secondary action appearing only when (1) the module is expanded AND (2) the expected output file exists in the working directory. The `checkOutputFile` function monitors output file existence through folder change events, automatically revealing visualization capabilities once results are available. This design eliminates user need to remember analysis execution status; the interface reflects data availability.

Visual hierarchy reinforces workflow: Visualize button appears above Feature Extract with white background and blue border indicating secondary role. Both implement hover detection with pointer cursor changes.

File Validation and Error Handling

The `validateTargetFile` function verifies necessary data exists before execution. For analysis modules, this target is typically `data_ICApplied_clean.mat` from preprocessing. The validator iterates through folder contents with case-insensitive comparison, setting descriptive error messages if required files are missing. An inline error text component displays messages in red, positioned near relevant controls.

Advanced Features

Edit Mode: When `editModeEnabled` is true, parameter controls expose additional functionality including renaming, syntax editing, and deletion. This transforms the template from consumption to configuration tool, enabling experienced users to customize analyses without leaving the application.

Integration: The template connects to parent page components and backend through property bindings and signal emissions. Properties like `currentFolder` and `folderContents` bind to shared application state tracking file browser location. Properties `moduleName`, `outputFileName`, and `visualizerFunction` parameterize each instance for its specific analysis type. Emitted signals `buttonClicked` and `visualizeClicked` trigger MATLAB executor calls with current parameters.

Extensibility: The template aliases the contentContainer's data property as default, allowing parent code to inject custom content into the expanded area, supporting future requirements

like module-specific help text or specialized controls without template modifications.

Z-index Management: Explicit z-index values of 1000 ensure expanded modules and controls appear above collapsed modules in stacked scenarios.

The template abstracts configuration and execution complexity behind an intuitive expand-collapse-configure-execute pattern that works uniformly across all analysis types. Parameter discovery, interface generation, validation, state tracking, and backend communication are handled automatically, accelerating development of new modules while ensuring improvements benefit all analysis types simultaneously.

Classifier Template

The Classifier Template orchestrates the complete machine learning workflow for training, configuring, and testing deep learning classifiers across multiple analysis types and architectures (EEGNet, EEG-Inception, Riemannian geometry-based approaches). Unlike the Analysis Module Template, it provides state management, persistent configuration storage, real-time logging, and testing capabilities within a unified interface.

Core Architecture

The template uses the collapsible panel design pattern, expanding to reveal analysis selection controls, time window sliders, classifier-specific parameters, class selection menus, action buttons, and logging displays. Dynamic height adjustment accommodates varying parameter counts, ensuring smooth layout transitions. Key properties:

- `classifierName`: Identifies classifier architecture; determines configuration files and backend functions
- `selectedAnalysis`: Tracks chosen analysis type (ERP, spectral, time-frequency, connectivity, intertrial coherence)
- `configParameters`: JavaScript object containing hyperparameters (layer dimensions, regularization, epochs, batch sizes) loaded from JSON and rendered via input boxes with immediate persistence

- `analysisTimeSelections`: Dictionary mapping classifier-analysis pairs to user-selected time windows; persists across sessions and application restarts

Time Window Management

Different analysis types require different temporal ranges due to distinct neuroscientific patterns. The `analysisTimeRanges` property stores four-element arrays [defaultStart, defaultEnd, minTime, maxTime] for each analysis:

- `ERP Analysis`: Default -0.2 to 1.0s; bounds -1.0 to 2.0s (baseline and post-stimulus event-related potentials)
- `Spectral Analysis`: Default 0 to 1.0s; bounds -0.5 to 3.0s (post-stimulus power changes)
- `Time-Frequency Analysis`: Default -0.2 to 1.0s; bounds -2.0 to 2.0s (extended baseline for spectral dynamics)

The `timeWindowProgrammaticUpdate` flag prevents circular update loops by distinguishing user-driven slider changes from programmatic updates, using Qt's `callLater` to reset the flag asynchronously. The `applyTimeWindowRange` function synchronizes both local properties and slider positions while managing persistence through the `savePersistedTimeSelections` method.

Analysis Selection and Configuration Loading

When users select an analysis from the dropdown, `loadConfigurationForAnalysis` orchestrates a cascading sequence:

1. Reload persisted time selections from disk (critical for multi-user/multi-instance concurrent access)
2. Query classifier controller via `loadAvailableClasses` for distinct class labels in selected analysis's feature data
3. Request classifier-specific parameters through `getParamsForAnalysis` (returns JSON with all hyperparameters for this classifier-analysis combination)

4. Initialize time window bounds and slider positions based on saved selections or appropriate defaults

This ensures all interface elements remain synchronized with the selected analysis type, preventing stale parameter persistence from previous selections.

Dynamic Parameters and Class Selection

A Repeater iterates over `configParameters` keys, instantiating InputBoxTemplate for each parameter. Type detection enables appropriate validation (numeric vs. textual). The `onValueChanged` handler implements immediate persistence: updates the in-memory object, invokes the controller's `saveConfiguration` method, and ensures configuration files always reflect interface state.

Class selection uses a split-button design with two sections: the main classify action and a right-side arrow toggle. The dropdown menu reveals checkboxes for each available class mapped to `selectedClasses` object entries. JavaScript immutability requires creating new objects with copied values when toggling selections to trigger QML property change notifications.

Training Workflow and Logging

The classify button validates input (analysis selected; at least one class chosen), then initializes training:

- Clears previous logs; sets `classifyExecuted` flag to true
- Populates initial log entries with timestamp-prefixed messages documenting configuration
- Emits `classifyClicked` signal carrying classifier name, analysis name, selected classes array, and complete `configParameters` object

A dedicated Window component displays a dark-themed console with monospaced text, automatically scrolling to show recent messages. The Connections component registers a

logReceived handler that checks the classifyExecuted flag and appends timestamped messages from the Python backend. Title dynamically incorporates classifier name for context during concurrent operations. An ”Open Console” button allows users to show the window after closing without restarting classification.

Weights Detection and Auto-Selection

The checkWeightsFileExists function monitors folder contents for files matching the pattern **[classifierName]_[analysisKey]_weights_best.keras**, checking for substring presence to accommodate qualifiers. When detected, the weightsFileExists property enables interface elements depending on trained model availability, and getWeightsFilePath constructs the absolute path.

The tryAutoSelectAnalysisFromWeights function automatically expands the template and selects an analysis if the current directory contains matching weights files for any supported analysis, activating only when users have not manually selected an analysis. This streamlines workflows for resuming training sessions.

Time Range Integration and Testing

The ”Select Time Range” button invokes the MATLAB executor's runSelectTimeRange method, triggering a MATLAB script that loads feature data, extracts temporal samples within the configured window bounds, and saves the windowed dataset. This enables experiments comparing classifier performance across different temporal windows.

A Test Classifier Window (currently disabled) provides a self-contained testing environment with properties for classifier name, selected analysis, weights file path, sample lists, and prediction results. The window connects to controller signals onSamplesLoaded and onSingleSampleResult, populates a dropdown with blind test samples (hiding ground truth labels), and displays predictions with visual hierarchy: predicted vs. actual labels in large boxes, correctness indicator (green for correct; red for incorrect), confidence percentage, and horizontal progress bars for class probability distributions. Dark-themed console at bottom displays test-specific logs with automatic scrolling.

Error Handling and Signal Communication

An inline `errorTextItem` (red text, positioned above action buttons) displays contextual error messages without modal dialogs, such as ”Please select an analysis type first” or ”Please select at least one class”. This follows best practices by positioning feedback adjacent to error sources.

The template emits three signals:

- `classifyClicked`: Complete classification request (classifier name, analysis name, selected classes, parameters)
- `testClassifierClicked`: Testing request (classifier name, analysis name, weights file path; currently inactive)
- `classificationComplete`: Results folder path for post-training navigation or visualization workflows

Signal-based architecture enables loose coupling between template and parent page implementations.

Advanced Features

Default Property Alias: Theoretically enables injection of custom content into the expanded area via `contentContainer`’s data property, accommodating future classifier-specific elements (training progress curves, configuration wizards, documentation panels) without template modifications.

Z-index Management: Root template item (z-index 1000) and classify dropdown menu (z-index 1001) ensure proper visual stacking when expanded. Essential for scenarios with multiple classifier templates preventing dropdown portions from rendering behind higher templates.

The Classifier Template abstracts parameter loading, configuration persistence, class selection, weights detection, and execution logging behind a consistent, intuitive interface regardless of classifier architecture or analysis type, enabling users to efficiently configure and monitor classification workflows.

II.6.2 MATLAB Parameter Parser & Dynamic Parameter Loader

MATLAB Parameter Parser

The parser in `src/parser/matlab_parameter_parser.py` scans MATLAB scripts for `cfg.*` assignments, classifies each match, and normalizes the result into a typed dictionary. Dedicated regular expressions capture ranges, start–step–end triplets, scalars, strings (with boolean coercion), colon or bracket arrays, and cell arrays that become multi-select string options. Matches are processed in file order with a small priority scheme so overlapping expressions resolve predictably, while internal helpers such as `ft_paths` are skipped. Each parsed entry records its inferred type plus sensible defaults (e.g., latency values gain a millisecond unit). The `ModuleParameterMapper` ties user-facing module names to one or more MATLAB files so multi-file modules merge their parameters. `DropdownOptionStore` loads curated dropdown metadata from JSON, lets users add or remove options, and persists updated bounds when sliders expand their ranges. Finally, `create_ui_component` combines the parsed value and curated metadata into a UI-ready spec (component type, labels, defaults, and selection limits) that the QML layer can render directly.

Dynamic Parameter Loader

The loader bridges the parsed parameters to the interface. `src/parser/dynamic_parameter_loader.py` resolves module file paths, parses them, enriches each parameter with stored dropdown metadata, and emits a JSON map consumed by QML. On the front end, `features/analysis/qml/DynamicParam` picks the correct control (range slider, stepped slider, dropdown, input box, checkbox), binds change signals to a globally exposed `matlabExecutor`, and mirrors edits back into MATLAB scripts. Range sliders push numeric spans, dropdowns support single or multi-select with custom option persistence, and input or checkbox updates serialize to MATLAB assignments. Helper logic keeps dropdown icons aligned inside nested layouts, exposes getters and setters for restoring state, and honors an edit mode to toggle between interactive and read-only views. Together, the Python loader and QML component keep the GUI and MATLAB configuration synchronized without manual rewrites.

II.6.3 Matlab Executor

The MATLAB Executor serves as the central bridge between the Python-based graphical interface and the MATLAB computational backend, orchestrating the entire execution lifecycle of neuroscientific analyses. This component embodies one of the most technically challenging aspects of NeuroPAC's architecture, managing cross-language communication, asynchronous process execution, file system operations, and dynamic configuration management across multiple analysis modules. The implementation resides within the `matlab_executor.py` file and comprises approximately five thousand lines of carefully structured code that handle dozens of interconnected responsibilities.

Architectural Foundation and Core Responsibilities

The MATLAB Executor is implemented as a class named `MatlabExecutor` that inherits from Qt's `QObject` base class, establishing it as a full participant in Qt's signal-slot communication mechanism. This inheritance enables seamless integration with the QML-based user interface through reactive property bindings and asynchronous signal emissions. Upon instantiation, the executor initializes several critical internal states including the current data directory path, references to various MATLAB script locations, and instances of helper objects such as the parameter parser and preprocessing bridge. The constructor also discovers and stores the project root directory, enabling consistent path resolution across different deployment scenarios whether the application runs in development mode or as a compiled executable through PyInstaller.

The class defines multiple PyQt signals that enable asynchronous communication with the user interface without blocking the main thread. The `outputChanged` signal delivers execution results to display panels, `configSaved` confirms successful parameter updates with detailed messages, `fileExplorerRefresh` triggers re-scanning of the file browser after data transformations, `processingFinished` notifies the interface when long-running operations complete, and `scriptFinished` returns results from background script executions. This signal architecture ensures that the interface remains responsive during computationally intensive MATLAB operations that might otherwise freeze the application.

Asynchronous MATLAB Execution Framework

One of the executor's most sophisticated features is its background thread management system for running MATLAB processes without blocking the user interface. The implementation employs a dedicated worker thread class called `MatlabWorkerThread` that executes MATLAB commands in isolated processes while the main application continues handling user interactions. When preprocessing or analysis operations are initiated, the executor spawns a worker thread configured with the MATLAB installation path, target script directory, and console visibility preferences. The worker constructs appropriate subprocess commands using either batch mode for headless execution or desktop mode when user interaction with the MATLAB console is desired.

The worker thread's execution logic handles numerous edge cases and error conditions. It normalizes file paths to use forward slashes for MATLAB compatibility regardless of the Windows-native backslash convention, wraps MATLAB commands in try-catch blocks to prevent silent failures, and implements comprehensive timeout mechanisms to prevent indefinitely hanging processes. When MATLAB completes execution, the worker emits a `finished` signal carrying a result dictionary that includes the return code, standard output, standard error, and any timeout indicators. The main executor receives this signal through its `_onMatlabFinished` callback method, which interprets the results, extracts relevant information such as the number of processed files, and generates appropriate user-facing messages before emitting them through the `configSaved` signal.

This asynchronous architecture proves essential for operations like Independent Component Analysis that can require several minutes to complete. Without background threading, the entire application would freeze during processing, creating a perception of unresponsiveness or failure. The worker thread pattern ensures that users can continue interacting with the interface, viewing documentation, or preparing subsequent analysis steps while computationally intensive operations proceed in the background.

Configuration Persistence and File Manipulation

The executor implements an extensive system for persisting user-configured parameters to MATLAB script files, ensuring that interface changes directly modify the underlying analysis code. This bidirectional synchronization maintains consistency between what users see in the GUI and what MATLAB executes, eliminating common sources of confusion where interface settings appear ignored because they were never actually written to the scripts. The persistence system operates through a family of slot methods decorated with `@pyqtSlot` that accept parameters from QML interface components and translate them into MATLAB syntax.

When a user adjusts a parameter in the interface, the corresponding QML component invokes one of several specialized persistence methods depending on the parameter type. The `saveDropdownPropertyToMatlab` method handles selection-based parameters, formatting single values as quoted strings or multiple values as MATLAB cell arrays. The `saveRangeSliderPropertyToMa` method processes numeric ranges, generating either bracketed array syntax for discrete intervals or colon notation for continuous ranges depending on the parameter's semantic meaning. The `saveCheckboxPropertyToMatlab` method converts boolean states into MATLAB's yes-no string convention, with special handling for interdependent parameters like demean and baseline window that must be enabled or disabled together.

The actual file manipulation occurs through a sophisticated pattern-matching and replacement system implemented in helper methods. The `_replace_or_insert_matlab_assignment` method uses regular expressions to locate existing parameter assignments within MATLAB scripts, preserving comments and maintaining consistent formatting while updating only the assignment value. If no existing assignment is found, the method intelligently determines an appropriate insertion point, typically placing new assignments before key function calls like `ft_preprocessing`. This approach ensures that dynamically added parameters integrate seamlessly with hand-written code rather than being appended haphazardly to file ends where they might be ignored or cause errors.

Special cases receive dedicated handling to manage complex interdependencies. The demean parameter in preprocessing, when disabled, must also comment out the baseline window specification since demeaning with a baseline window is meaningless. The `_save_demean_with_baseline`

method implements this logic by coordinating modifications to both properties, using `_comment_matlab_assi` and `_uncomment_matlab_assignment` helpers to preserve the baseline window values in commented form rather than deleting them entirely. This preservation allows users to re-enable demeaning with their previously configured baseline without needing to remember and re-enter the values.

Dynamic Parameter Reading and Interface Initialization

Complementing the configuration writing system is an equally comprehensive parameter reading framework that initializes the interface with current values from MATLAB scripts when modules are loaded. This capability ensures that the GUI always reflects the actual state of the analysis configuration rather than defaulting to arbitrary initial values that might not match the script contents. The executor provides numerous slot methods prefixed with `getCurrent` that extract specific parameter values through targeted regular expression searches.

Methods like `getCurrentPrestim`, `getCurrentPoststim`, and `getCurrentEventvalue` parse preprocessing script contents to retrieve trial definition parameters. The `getCurrentBaselineWindow` method demonstrates sophisticated error handling by first searching for uncommented baseline window specifications and falling back to commented versions if no active specification exists, ensuring that users can see their baseline configuration even when demean is currently disabled. The `getCurrentChannels` method reads channel selection arrays from the preprocessing module, handling variations in whitespace and formatting that might arise from manual script editing.

These reading methods employ defensive programming practices that gracefully handle missing or malformed parameters. Each method wraps its file operations and parsing logic in exception handlers that log detailed error messages for debugging while returning sensible default values to prevent interface initialization failures. This robustness ensures that the application remains functional even when working with partially configured or manually modified scripts that might not conform to expected patterns.

The parameter reading system integrates with the dynamic parameter loader during module initialization. When a user selects an analysis module, the executor's `getModuleParameters`

method orchestrates the discovery and extraction of all parameters from the module's associated MATLAB files. This method delegates file parsing to the `MatlabParameterParser` class, aggregates parameters from multiple script files when modules span several files, updates range slider bounds based on current values to ensure sliders can display and modify existing settings, and generates JSON-formatted component specifications that the QML interface uses to construct appropriate input controls.

Path Resolution and Cross-Platform Compatibility

Managing file paths correctly across Python, MATLAB, QML, and the Windows operating system presents significant challenges that the executor addresses through systematic normalization and conversion. The executor maintains multiple helper methods dedicated to path handling, ensuring that paths remain valid as they traverse different runtime environments. The `resource_path` function resolves relative paths to absolute paths that work both during development and in PyInstaller-packaged executables where resources are extracted to temporary directories.

When receiving paths from QML, the executor strips the `file:///` URL scheme prefix that Qt's file dialogs prepend, converts forward slashes to backslashes for Windows path operations, and normalizes multiple backslashes that might accumulate during conversions. Before passing paths to MATLAB, the executor performs the reverse transformation, converting backslashes to forward slashes since MATLAB interprets backslashes as escape characters in string literals. This bidirectional path normalization occurs in methods like `updateDataDirectory` and `updateFieldtripPath`, ensuring consistent behavior regardless of how users select directories.

The executor also implements intelligent path resolution for locating MATLAB script files across the project structure. Methods like `_get_preprocess_data_script_path`, `_get_timelock_script_path` and similar functions maintain lists of candidate paths where scripts might reside, checking each location in order of likelihood and returning the first valid match. This approach accommodates different installation scenarios, allows for restructured project layouts during development, and provides fallback paths for PyInstaller-bundled resources that might be extracted to

non-standard locations.

Custom Component Management

The executor implements a sophisticated system for managing dynamically created interface components that users add through the GUI, enabling extensibility beyond the predefined parameter set. This functionality supports custom dropdown menus and range sliders that users can create to expose additional MATLAB configuration options without requiring code modifications. The implementation involves direct manipulation of QML source files, inserting properly formatted component definitions that persist across application restarts.

When a user requests a custom dropdown through the interface, the `addCustomDropdownOption` method receives parameters specifying the dropdown's label, MATLAB property binding, multi-select behavior, and initial item lists. The method generates a complete QML component definition using the `_build_custom_dropdown_snippet` helper, which constructs a properly indented and escaped `DropdownTemplate` block with all necessary properties including persistence identifiers, label text, option arrays, and selection limits. The insertion logic locates a designated container element within the QML file using `_locate_custom_container_bounds`, finds the container's closing brace through brace-matching depth tracking, and splices the new component definition into the file content just before the closing brace.

Similar functionality exists for custom range sliders through methods like `addCustomRangeSliderOption` and `_build_custom_range_slider_snippet`. These methods generate `RangeSliderTemplate` definitions with numeric bounds, step sizes, units, and current values. The executor assigns each custom component a unique identifier following a sequential numbering scheme, determined by scanning existing component IDs and incrementing from the maximum found index. This ensures that multiple custom components coexist without identifier collisions.

The executor also provides removal and modification operations for custom components. The `deleteCustomDropdown` and `deleteCustomRangeSlider` methods locate component blocks using regular expression searches, remove them from the QML content, and rewrite the file. Update operations employ a replace strategy, generating a new component definition with modified parameters and substituting it for the old block while preserving the compo-

nent's identifier. These manipulation capabilities transform the interface from a static parameter display into a living configuration system that users can adapt to their specific analysis requirements.

Module-Specific Targeting and Script Selection

A critical aspect of the executor's design is its ability to route parameter updates to the correct MATLAB script files based on the active analysis module. Different modules employ different script files located in distinct directories, and parameters with identical names in different modules must modify different files to avoid cross-contamination of configurations. The executor implements a module-aware targeting system that maps module names to their corresponding script paths and ensures parameters reach their intended destinations.

The core of this system resides in conditional logic within persistence methods that examines the `module_name` parameter to determine appropriate target scripts. For instance, `saveDropdownPropertyToMatlab` contains branching logic that assigns the preprocessing data script when `module_name` equals "Preprocessing", the spectral analysis script for "Spectral Analysis", the time-frequency script for "Time-Frequency Analysis", and the timelock script for "ERP Analysis". Each branch constructs a list of target scripts, and subsequent code iterates through this list performing the same update operation on each file.

This architecture supports scenarios where a single parameter might legitimately need to appear in multiple scripts. For example, channel selection parameters could apply to both preprocessing and analysis stages, requiring updates to scripts in both modules. The executor handles such cases by populating the target script list with multiple paths when appropriate, then iterating through all targets to ensure consistent configuration across the analysis pipeline.

The module targeting system also implements fallback logic for backward compatibility and error recovery. When an explicit module name is not provided, the executor attempts to infer the appropriate target based on the parameter name itself. Properties like `cfg.foi` and `cfg.taper` strongly suggest spectral analysis, while `cfg.toi` and `cfg.width` indicate time-frequency analysis. This inference allows legacy interface code that predates the explicit module naming convention to continue functioning correctly. However, the executor prints warning

messages when relying on inference, encouraging developers to provide explicit module names for clarity and preventing ambiguous cases where inference might select incorrect targets.

Data Formatting and MATLAB Syntax Generation

Translating user interface values into valid MATLAB syntax requires careful attention to numerous formatting conventions and data type distinctions. The executor implements a family of formatting helper methods that generate syntactically correct MATLAB code from various Python data structures. These formatters handle the impedance mismatch between Python's dynamic typing and MATLAB's specific syntax requirements for different data types.

The `_format_matlab_assignment_value` method serves as the primary string and array formatter. It receives a list of values and a boolean indicating whether cell array format is required, then generates either a single quoted string, a numeric literal, a bracketed numeric array, or a cell array with curly braces depending on the input characteristics. Single values that parse as numbers are rendered as numeric literals without quotes, while non-numeric values receive single-quote wrapping with proper escaping of embedded quotes using MATLAB's double-quote escape convention. Multiple values are formatted as cell arrays with space-separated quoted elements, ensuring MATLAB interprets them as string cell arrays rather than character arrays.

Numeric range formatting employs specialized methods for different MATLAB range representations. The `_format_matlab_numeric_range` method generates two-element bracketed arrays for discrete ranges like frequency bands or time windows. The `_format_matlab_colon_range` method produces colon notation for sequences specifying start, step, and end values such as time-of-interest vectors in time-frequency analysis. The formatter intelligently strips unnecessary trailing zeros and decimal points from floating-point representations, rendering "5" instead of "5.000000" for integer-valued floats while preserving meaningful decimal places for non-integer values.

Boolean parameters receive special handling through the checkbox persistence logic, which converts Python boolean values to MATLAB's string-based yes-no convention. This conversion recognizes that MATLAB configuration parameters typically expect `'yes'` or `'no'` strings

rather than logical true-false values, reflecting FieldTrip's configuration style. The formatter wraps these strings in single quotes and inserts them into assignments that MATLAB's preprocessor can correctly interpret.

The formatting system also manages comment preservation during parameter updates. When replacing parameter assignments, the executor uses regular expressions that capture any trailing comments on the assignment line, preserving them in the updated line to maintain documentation and user notes embedded in the scripts. This attention to detail prevents the automated updates from destroying valuable human-readable information that helps users understand their configuration choices.

Classification Pipeline Integration

Beyond preprocessing and feature extraction, the executor provides specialized support for the classification module, handling data transformation and time window selection operations that bridge MATLAB feature extraction with Python machine learning frameworks. The classification integration relies heavily on the `PreprocessBridge` class, which the executor instantiates during initialization with the current data directory path. This bridge object abstracts the complexity of loading MATLAB-generated feature matrices and transforming them into NumPy arrays suitable for scikit-learn and TensorFlow models.

The `loadAndTransformData` slot method exemplifies this integration, accepting an analysis type specification and target model name from the classification interface. The method delegates to the preprocessing bridge's `load_and_transform` function, which loads appropriate `.mat` files from the data directory, extracts relevant fields based on the analysis type such as ERP amplitudes or spectral power values, reshapes multidimensional arrays into feature vectors compatible with classifier input requirements, and returns a NumPy array ready for training. The executor wraps this process in comprehensive exception handling, capturing file-not-found errors when data has not been generated, key errors when analysis types are misspelled or data structures do not match expected formats, and generic exceptions for unexpected failures, returning descriptive JSON error messages to the interface for user feedback.

The time window selection functionality, implemented through `runSelectTimeRange` and

its worker thread `_run_select_time_range_worker`, enables classifiers to focus on specific temporal regions of interest within trials. When invoked, the executor launches a background thread that constructs a MATLAB command calling the `select_time_range.m` script with classifier name, analysis type, data path, and a JSON configuration file specifying window boundaries. The MATLAB script opens in desktop mode, allowing users to interactively inspect the data and refine their selections if needed. Upon completion, the script writes time-windowed data to a new output folder with a `_timeranged` suffix, and the executor emits signals to refresh the file browser so users can immediately locate and utilize the transformed data.

This classification support demonstrates the executor's role as an integration hub, coordinating multiple technologies and frameworks to present users with a unified workflow despite the underlying heterogeneity of computational tools.

Error Handling and User Feedback

Throughout its extensive functionality, the executor implements comprehensive error handling and user feedback mechanisms that transform raw exceptions and failures into actionable information. Nearly every public slot method wraps its core logic in try-except blocks that catch anticipated exceptions, log detailed diagnostic information to the console for debugging, and emit user-friendly messages through the `configSaved` signal that appear in the interface's status display. This approach ensures that errors do not crash the application or leave it in undefined states, instead providing users with clear descriptions of what went wrong and, when possible, suggestions for resolution.

Error messages distinguish between different failure modes with appropriate specificity. File-not-found errors clearly state which file is missing and suggest prerequisite operations that should be completed first, such as running preprocessing before attempting ICA component visualization. Parse errors in parameter reading methods indicate which parameter could not be extracted and fall back to reasonable defaults, allowing the interface to initialize in a partially configured state rather than failing entirely. Subprocess execution failures report MATLAB return codes and include both standard output and standard error from the MATLAB process,

giving users visibility into MATLAB's own diagnostic messages.

The executor also implements graceful degradation strategies for non-critical failures. When custom component manipulations fail due to unexpected QML file structures, the executor logs warnings but does not prevent normal operation since custom components are optional enhancements rather than essential features. When parameter writes fail for one target script in a multi-target update, the executor continues updating remaining targets and reports a summary of successes and failures rather than aborting the entire operation after the first error.

This defensive programming philosophy, combined with extensive logging and transparent error reporting, transforms the executor from a brittle integration point that could easily fail into a robust coordination layer that maintains operational stability even when confronting unexpected conditions.

State Management and Configuration Synchronization

The executor maintains several pieces of internal state that track the application's configuration and execution status, ensuring consistency across asynchronous operations and multiple interface interactions. The `_current_data_dir` property stores the actively selected data directory, initialized by reading the preprocessing script at startup and updated whenever users select new folders through the file browser. This cached value enables quick reference without repeatedly parsing scripts, and the executor propagates updates to dependent objects like the preprocessing bridge whenever the data directory changes.

The `_option_store` instance, a `DropdownOptionStore` object, maintains persistent storage of custom dropdown options and range slider bounds across application sessions. This store serializes to a JSON configuration file, and the executor invokes its save method whenever custom options are added or range limits are updated. During module parameter discovery, the executor queries the option store to retrieve saved configurations for each parameter, merging them with current values from MATLAB scripts to produce complete component specifications. This integration ensures that dynamically added options persist across application restarts and remain synchronized with script-based configurations.

Thread tracking through properties like `_worker_thread` and `_select_time_thread` pre-

vents concurrent execution of operations that should not overlap. Before launching new background processes, the executor checks whether existing threads are still alive and emits warning messages if users attempt to start duplicate operations. This safeguard prevents resource exhaustion from accumulating subprocess instances and avoids race conditions where multiple threads might simultaneously modify the same files.

The executor's state management embodies a consistent philosophy of single-source-of-truth, where MATLAB script files serve as the authoritative configuration source, the Python executor maintains runtime state derived from those files, and interface components reflect the executor's state through reactive property bindings. This unidirectional data flow pattern, inspired by modern frontend frameworks, eliminates many categories of synchronization bugs and makes the application's behavior more predictable and maintainable.

II.6.4 File Browser Window

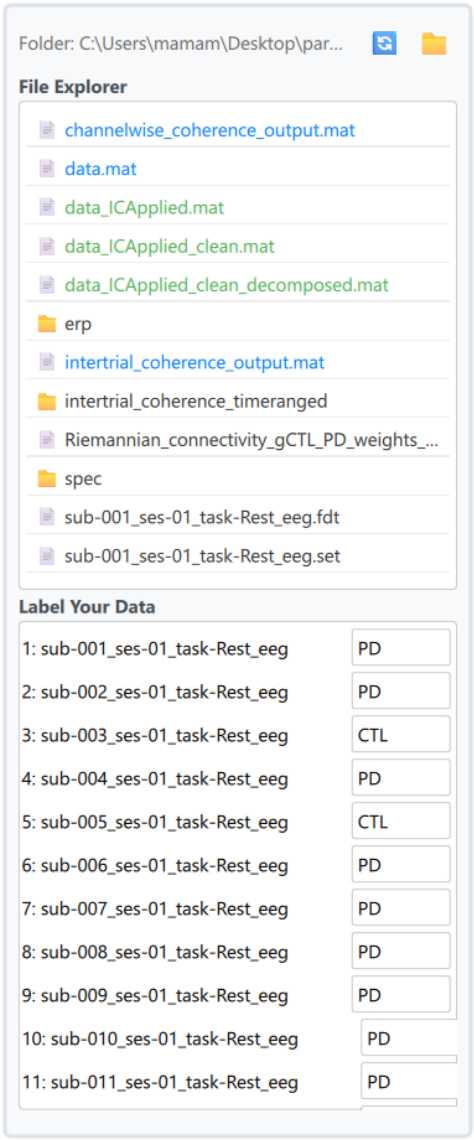


Figure II.11: File Browser Interface displays the file explorer with folder contents, the lower section provides data labeling for annotating selected dataset.

The file browser serves as the primary interface for navigating the file system, selecting datasets, and labeling experimental data. It bridges the QML frontend with a Python backend using a reactive architecture: a `FileBrowser` class (inheriting from `QObject`) manages file operations, while `FileBrowserUI` QML component maintains signal-slot connections that ensure the interface always reflects backend state.

Backend Implementation and Path Normalization

The Python `FileBrowser` class maintains three internal states exposed through Qt proper-

ties: current folder path, folder contents list, and processed file data in RAM. Property changes emit `pyqtSignal` events that trigger QML updates without explicit method calls, eliminating synchronization issues.

The `loadFolder` method handles cross-platform path complexity through defensive normalization:

1. Receives paths from Qt's file dialog (format: `file:///C:/Users/...`)
2. Strips `file:///` prefix and removes double backslashes
3. Converts forward slashes to backslashes for Windows `os` module compatibility
4. Uses `os.listdir()` to enumerate items; `os.path.isdir()` determines type
5. Prefixes files with document symbol, directories with folder symbol
6. Emits `folderContentsChanged` signal with formatted items; catches exceptions (inaccessible paths, permission errors) and reports to user

This robust approach ensures consistent path handling regardless of source (file dialogs, MATLAB scripts, user input).

Frontend Architecture and Visual Layout

The `FileBrowserUI` component uses Qt's `Connections` element to register handlers for backend signals, exemplifying the model-view-controller pattern: Python backend (model/logic), QML component (view/controller). The layout occupies the left 20

- **Upper region**: File explorer displaying folder contents with scrollable list view
- **Lower region**: Data labeling interface for annotating datasets

Header bar (spanning both regions) displays the current folder path with two buttons:

- `Refresh` button: Reloads folder contents without changing directories
- `Folder Selection` button: Opens native file picker

Responsive height computation allocates screen space dynamically, maintaining minimum heights to ensure usability. Computed properties calculate available space and distribute fairly between explorer and labeling sections.

File Interaction and Type-Specific Handling

The file explorer renders items as interactive elements with hover effects. Hover state displays light blue background; MATLAB files (`.mat`) gain underline styling. Color coding distinguishes file types:

- `.mat` files: Blue text (indicating interactivity)
- ICA component files (filename contains "ICA"): Green text with distinctive icon

Click handling implements type-specific behavior:

- **data.mat files**: Emit `fileMatClicked` signal; parent component populates labeling interface with file contents
- **ICA component files**: Delegate to MATLAB executor, launching FieldTrip's interactive `ft_databrowser` for component inspection and rejection
- **Other files**: Simple notification signal (no further action)

Data Labeling Interface

When a `data.mat` file is selected, the system queries the Python backend through `matlabExecutor` to retrieve dataset names (MATLAB `.mat` files can contain multiple named arrays and structures). The backend:

- Returns dataset name list
- Loads previously saved labels from persistent storage
- Populates list model with dataset-label pairs

The interface presents pairs as input fields bound to the list model through QML property binding, automatically persisting user-entered label changes. When users press enter or move focus away from a field, labels save to Python storage, establishing permanent dataset-semantic associations.

Advanced Features

Dual-Pane Design: File explorer and labeling interface in stacked layout keeps frequently accessed controls visible while providing specialized functionality below. This thoughtful UX maintains immediate access to core operations while organizing secondary features.

External Refresh: The `refreshFromExternal()` function provides a public entry point for parent components to reload folder contents without requiring user interaction. Enables seamless workflows where classification results or analysis outputs automatically appear in the browser immediately after computation, without manual refresh clicks.

The file browser abstracts file system complexity and persistent storage behind a clean interface, enabling users to efficiently organize and annotate datasets for subsequent analysis modules.

II.6.5 Preprocessing-Specific

ICA Browser

The ICA Browser enables inspection and rejection of independent components extracted from EEG data, bridging automated decomposition with human judgment for identifying physiological artifacts (eye blinks, muscle activity, cardiac signals).

Data Loading and Variable Detection

The browser loads preprocessed ICA data from a specified MATLAB file path, performing intelligent variable detection by examining all non-metadata variables. The detection algorithm prioritizes `data_ICApplied` (conventional ICA decomposition container) but flexibly identifies ICA data structures through characteristic FieldTrip properties: channel labels, trial matrices, temporal information arrays. This structural validation enables operation on diverse

preprocessing outputs without rigid naming conventions.

Sensor-Space Data Handling

The system attempts to locate a companion `data.mat` file containing raw sensor-level measurements. If found, loads directly. Otherwise, implements a fallback mechanism reconstructing sensor-space data from ICA components via matrix multiplication: applies topographical transformation matrices to project independent components back to original electrode space. Reconstruction preserves metadata (sampling rates, trial information, recording headers) by traversing FieldTrip's configuration history chain.

Interactive Review and Batch Processing

The browser processes subjects sequentially, presenting FieldTrip's `ft_databrowser` in component view mode. This displays each component's temporal evolution alongside its topographical scalp distribution, enabling artifact identification through time-series and spatial signatures. Rejection decisions accumulate in a structured array maintaining subject-component correspondence.

After interactive review, an automated pipeline applies rejection decisions in batches (10 subjects at a time for memory efficiency). For each subject, reconstructs channel signals from only retained components, filtering artifact-related neural activity while preserving cognitive signals. Batch processing with incremental disk persistence protects against data loss and allows interrupted sessions to resume from last successful batch.

Stimulus-Specific Decomposition and Outputs

Following artifact removal, the browser segments each cleaned dataset into three condition categories (target, standard, novelty stimuli) by examining trial information codes. Prepares data for condition-specific analyses (ERP averaging, spectral power comparisons).

Three primary outputs:

- **Cleaned data**: Complete temporal/spatial structure with artifacts removed
- **Decomposed data**: Experimental conditions separated into discrete structures

- **Rejection records**: Audit trail documenting which components were rejected per subject for quality control

All outputs persist to disk in MATLAB's compressed format and export to workspace for immediate validation.

Reject IC Button & `ft_databrowser` Synchronization

Integrating NeuroPAC's ICA workflow with FieldTrip's `ft_databrowser` required a strategic modification approach extending FieldTrip capabilities while maintaining compatibility with future updates and minimizing disruption to users' existing installations.

Architecture: Minimally Invasive Modification

Rather than forking FieldTrip entirely (creating maintenance burden and breaking shared installations), the solution adopts a minimally invasive approach modifying only ICA component rejection functionality. Two parallel versions exist:

- **Reference version**: Stored in NeuroPAC at `features/preprocessing/matlab/ft_databrowser_mo` containing all ICA rejection modifications
- **User's installation version**: Located in user's FieldTrip directory, receives modifications through automated synchronization

Core Modification: Reject IC Button

The modification introduces a "reject IC" button in `ft_databrowser`'s right column control panel using normalized coordinates for resolution independence. Button instantiation via `uicontrol` specifies parent figure handle, pushbutton style, and unique user data identifier 'rc' linking to callback handler.

When clicked, the interface enters interactive selection mode. The callback implements position-to-component mapping by analyzing spatial layout and determining which timecourse is nearest to click coordinates. The algorithm differs by view mode:

- **Butterfly view**: All components overlay in single axis

- **Vertical/component view**: Each element occupies dedicated spatial region; directly queries figure's timecourse objects using `findobj` with 'timecourse' tag, exposing y-data through graphics handles. Identifies temporal sample nearest to horizontal click coordinate, then determines component by finding which timecourse has nearest y-value at that time sample. This reverse approach (rendered graphics → data indices) proves robust by implicitly accounting for all FieldTrip scaling/transformation operations

Rejection Data Structure and Visual Feedback

Once a component is identified, the system extracts component number/label from the data structure's channel label array. Accesses `cfg.current_subject_index` passed by `browse_ICA.m`, which iterates through subjects sequentially. Maintains cell array `opt.reject_ICs` (one cell per subject, initialized with zeros):

- Single rejection: Cell contains numeric scalar
- Multiple rejections: Transitions to nested cell array with numeric values

This hybrid structure balances memory efficiency with query simplicity.

Visual feedback highlights the selected timecourse in red with "(REJECTED)" label at click location, persisting until user selects different component or navigates to different trial/subject. Console prints diagnostic messages (component name, number, subject index) creating audit trail.

When databrowser closes, a cleanup callback packages state into output configuration, adding `cfg.rejected_ICs` field to standard FieldTrip outputs. This minimal semantic extension maintains backward compatibility.

Synchronization Mechanism

The Python script `quick_sync.py` propagates modifications from NeuroPAC's reference version to user's FieldTrip installation. Implements intelligent update system:

1. Loads preprocessing configuration to extract user's FieldTrip installation path
2. Compares modification timestamps of reference and installation versions

3. If reference newer: Synchronizes; if installation current/newer: Exits

4. Performs targeted cleanup removing standalone testing blocks (path setup commands)

5. Writes modified content to user's `ft_databrowser.m` via direct replacement

6. Includes comprehensive error handling (permission problems, missing files, unexpected structures)

Timestamp-based approach provides efficient change detection without expensive file content comparison.

Integrates as optional preliminary step users invoke through menu option before ICA analysis. Transparent instructions explain installation modification and recommend backing up original `ft_databrowser.m`. Users declining synchronization retain preprocessing features but lack integrated component rejection.

Communication Flow and Result Collection

`browse_ICA.m` constructs configuration structures with standard FieldTrip parameters (layout specifications, view mode) plus NeuroPAC-specific context (`cfg.total_subjects`, `cfg.current_subje
After databrowser returns, extracts rejection information from `cfg.rejected_ICs` field, consolidating into master rejection array with defensive extraction logic handling various data structure possibilities (missing fields, empty arrays, nested cells, inconsistent indexing).

The consolidated array drives artifact removal: iteratively invokes FieldTrip's component rejection function per subject with appropriate rejection indices. Separation between interactive selection and batch processing allows continuous subject review without waiting for computations, enables decision revision before final processing, and facilitates batch optimization.

The architecture reflects key software engineering principles: minimal modification reduces upstream divergence; timestamp-based synchronization provides automatic updating; transparent user notification respects autonomy while streamlining setup.

II.6.6 Analysis-Specific

Analysis Visualizer

Shared interaction model

All visualizers follow a consistent MATLAB GUI pattern to keep analysis exploration predictable. Each tool: (i) normalizes inputs into a subjects-by-condition struct array (target/standard/novelty) to avoid dissimilar-structure errors; (ii) stores state in the figure via `guidata` (current subject, selected channels or pairs, axis overrides); (iii) provides Previous/Next navigation to iterate subjects; (iv) offers checkbox panels to select channels (or channel pairs) with an "all" toggle; (v) exposes `xlim`/`ylim` overrides through edit boxes that immediately re-render plots; and (vi) hides the figure until UI controls are fully built to avoid flicker. All UIs enforce defensive handling of empty or malformed inputs and clamp selections to valid indices.

FieldTrip dependency handling

The time–frequency and coherence visualizers attempt to auto-locate FieldTrip (e.g., checking `ft_singleplotTFR`/`ft_connectivityplot` availability) and mirror the fallback search logic used in preprocessing scripts. When available, FieldTrip plotting routines are invoked with interaction disabled (e.g., `cfg.interactive = 'no'`) so that NeuroPAC's own UI remains authoritative.

ERP visualizer (time-domain averages)

Displays per-channel event-related potentials in a channel-by-condition grid using `avg` waveforms. Typical use: rapidly compare latency/amplitude differences across target, standard, and novelty conditions with optional manual axis limits. Uses the recorded channel labels when present, falling back to autogenerated labels otherwise.

Spectral visualizer (frequency-domain power)

Plots power spectral density curves derived from `fourierspctrm` for each selected channel. Supports subject paging and channel selection identical to ERP, with optional frequency/Power axis overrides to focus on bands of interest. Generates default channel labels when none are

provided.

Time–frequency visualizer (TFR surfaces)

Renders time–frequency power maps per condition via FieldTrip's `ft_singleplotTFR`. Requires at least one channel selection; otherwise it suppresses plotting with a user message. Preserves provided `layout` when present, and applies optional time/frequency window overrides. Input normalization accepts either a subjects-by-3 matrix or a vector of structs with {target, standard, novelty} fields, copying only known fields to avoid structure mismatch.

Channel-wise coherence visualizer

Targets connectivity outputs (coherence) across channel pairs. Builds a pair list from `labelcmb` or square coherence matrices, then plots time–frequency coherence per selected pair across the three conditions. Provides pair-level checkboxes (two-column layout when many pairs exist) plus shared axis overrides. Extracts pair-specific slices into FieldTrip-compatible TFR structs for plotting.

Inter-trial coherence visualizer

Shows per-channel ITPC/ITLC maps, switchable via a coherence mode dropdown (phase or linear). Otherwise mirrors the ERP-style navigation and channel selection. Uses `itpc` or `itlc` fields when present, with graceful skips for missing data and optional axis limits to highlight time–frequency regions.

II.6.7 Classification-Specific

Preprocess Bridge

Purpose and role

The Python module `preprocess_bridge.py` bridges MATLAB analysis outputs and Python deep-learning models. It discovers and loads FieldTrip-derived `.mat` files (ERP, time–frequency, spectral, connectivity, inter-trial coherence), normalizes their structures, pads variable shapes, and returns tensors plus labels suitable for model input.

Core data structures

`MatlabStructElement` wraps MATLAB structs so they behave like Python dicts and attributes while preserving keys, shapes, and simple metadata. `PreprocessBridge` holds resolved data paths, filename mappings per analysis type, expected condition order (target, standard, novelty), and flexible group label mappings (e.g., HC/CTL vs. PD variants) sourced from a shared `labels` list.

Loading pipeline

`load_and_transform()` cleans incoming paths (stripping `file:///` prefixes, normalizing slashes), picks the right `.mat` filename, then loads with `scipy.io.loadmat`; MATLAB v7.3 files automatically fall back to `h5py`. It selects the correct struct key per analysis (e.g., `ERP_data` with `avg`, `timefreq_data` with `powspctrm`), iterates subjects in struct arrays, derives dataset names from `cfg.dataset` when available, and attaches group labels from `labels`. For each condition it retrieves the primary data field with fallbacks (trial averaging when needed), converts complex arrays to magnitudes, handles spectral/connectivity-specific averaging, cleans NaN/Inf values, and records condition and subject indices.

Shape normalization and transforms

After extraction, samples are squeezed, dimension-aligned by padding to the max observed shape, and stacked into X. Labels y include condition, group, subject id, and dataset name. `_apply_transform()` reshapes ERP data for EEGNet/EEG-Inception (adds singleton spatial dimension) and preserves 4D tensors for time–frequency and inter-trial coherence when targeting EEG-Inception, while passing through Riemannian targets untouched.

Filtering and selection

`filter_by_selected_classes()` supports class-specific downselection (e.g., only PD-target) by parsing class strings into group/condition pairs, normalizing group labels (numeric or textual), and returning filtered samples plus kept indices.

MATLAB v7.3 handling

Helper methods (_load_mat_v73, _convert_mat73_struct_array, _convert_condition_struct) dereference HDF5 references, extract only relevant data fields, and wrap them in MatlabStructElement, ensuring v7.3 files remain interchangeable with earlier loadmat outputs.

Controller

Role and architecture

The classification controller coordinates training, testing, sample loading, and single-sample inference through Qt signals/slots and worker objects that run in separate QThreads, keeping the UI responsive while streaming live log output from subprocesses.

Worker responsibilities

TrainingWorker, TestWorker, SampleLoaderWorker, and SingleSampleClassifierWorker each wrap a specific workflow. They resolve the model directory (EEGNet, EEG-Inception, Riemannian), build the correct script path (main.py, test.py, load_samples.py, classify_sample.py), launch a Python subprocess with the chosen analysis key and data path, stream stdout to the UI, and emit JSON-tagged results when present (e.g., JSON_RESULT, JSON_SAMPLES, JSON_CLASSIFICATION). Training additionally passes selected classes (as JSON) and optional config parameters; testing accepts weights paths; loaders focus on discovery without inference.

Controller duties

ClassificationController owns a ConfigParser for analysis/model metadata, tracks the active data folder, and exposes slots to QML for listing classifier configs, available analyses, and class combinations built from diagnosis_labels crossed with the canonical conditions (target, standard, novelty). It normalizes incoming paths from the UI (stripping file:///), prevents concurrent runs by checking an existing thread, wires worker signals to UI signals (logs, completion, results), and cleans thread references after completion.

Training and classification flow

startClassification() converts display names to analysis keys, validates that a data folder is set, logs selected class filters, spawns a TrainingWorker, and streams subprocess

output. Training optionally writes per-analysis JSON configs into the model's `configs` directory before execution. Similar patterns drive `startTraining()` when launched without UI-specified analysis/config overrides.

Testing and inference flow

`testClassifier()` normalizes weight file paths, resolves the analysis key, and runs a `TestWorker`; JSON results are forwarded via `testResults`. `loadTestSamples()` invokes `SampleLoaderWorker` to list available samples without classifying. `classifySingleSample()` runs `SingleSampleClassifierWorker` with a specific weight file and sample index, emitting a single-result JSON payload.

Persistence and utilities

The controller keeps a shared time-window selection file under `config/time_window_selections.json`, ensuring parent directories exist and providing load/save slots for UI persistence. Helper slots return diagnosis labels, condition labels, counts, and analysis-key mappings. Defensive try/except blocks surface errors to the UI via log signals instead of crashing the application.

Inference

Unified inference architecture

All three classification models (EEGNet, EEG-Inception, Riemannian) follow a consistent inference pattern with dedicated wrapper classes that isolate weight loading, preprocessing, and prediction into standardized interfaces. Each inference module exposes a class constructor accepting a weights path, a `preprocess()` method normalizing input tensors, and a `predict()` method returning class predictions plus probability distributions (when available). This uniform API enables the controller to treat models interchangeably at runtime, switching between deep learning and traditional machine learning approaches without code duplication.

Common inference logic

All implementations share core responsibilities: (i) weight loading with exception handling that surfaces file-not-found or corruption errors gracefully; (ii) batch dimension han-

dling to accept both single-trial (channels, samples) and multi-trial (trials, channels, samples) inputs, automatically expanding singleton dimensions; (iii) defensive validation ensuring loaded artifacts (Keras models or scikit-learn pipelines) are properly instantiated before inference; (iv) unified output format returning predicted class indices alongside probability vectors (or None when unavailable); and (v) scaling support via optional scale_factor parameters to convert units (e.g., volts to microvolts) without altering stored data.

Deep learning inference (EEGNet and EEG-Inception)

Both neural network models leverage TensorFlow/Keras backends and set K.set_image_data_format('c
to enforce consistent tensor ordering. Initialization requires architecture hyperparameters (number of classes, channel counts, temporal samples, filter configurations, dropout rates) mirroring those used during training; the constructor rebuilds the model graph then loads saved weights via model.load_weights(). Preprocessing applies data scaling first, expands single-trial inputs to batch form (1, channels, samples), then reshapes to match expected 4D tensor layouts—EEGNet expects (trials, channels, samples, 1) with a trailing singleton dimension representing convolutional kernels, while EEG-Inception transposes channel and time axes (trials, time, channels) before adding the singleton to yield (batch, time, channels, 1), reflecting its time-major architecture. Prediction invokes model.predict(), which returns softmax probabilities; class labels are extracted via argmax(-1).

Riemannian inference (pyRiemann pipeline)

The Riemannian approach loads a complete scikit-learn pipeline serialized with joblib, avoiding architecture reconstruction entirely. This pipeline encapsulates covariance estimation, tangent space projection, and SVM/MDM classification in a single artifact. Initialization only requires the .pkl path; validation confirms file existence before loading. Preprocessing is minimal—ensuring the input is 3D (trials, channels, features) by expanding 2D arrays—since feature extraction (spectral power or connectivity matrices) occurs upstream in the analysis bridge. Prediction calls pipeline.predict() directly; probability retrieval attempts predict_proba() but gracefully returns None when the final estimator lacks this method (e.g., Minimum Distance to Mean classifiers). Additional introspection meth-

ods (`get_pipeline_steps()`, `get_feature_importance()`) expose internal pipeline structure and SVM coefficients when available, supporting interpretability workflows absent from black-box neural models.

Shape handling divergences

Input tensor transformations differ per model's inductive biases. EEGNet's depthwise-separable convolutions operate channel-wise on temporal sequences, accepting `(trials, channels, samples, kernels)` with standard channel-first EEG ordering preserved except for the trailing dimension. EEG-Inception's multi-scale inception modules process time-first sequences `(batch, time, channels, 1)`, necessitating a transpose from the canonical `(batch, channels, time)` input; this reordering aligns temporal convolutions along axis 1. Riemannian methods expect pre-extracted features in `(trials, channels, features)` where features encode frequency-band power or connectivity coefficients, not raw time-series samples; the inference module assumes these features arrive already computed from upstream spectral or coherence analyses, avoiding redundant transformations.

Error handling and robustness

All classes wrap weight loading in try-except blocks, printing diagnostic messages before re-raising exceptions to aid debugging. Deep learning models catch `load_weights()` failures (missing files, incompatible architectures); Riemannian inference validates file existence via `os.path.exists()` prior to `joblib.load()`. Preprocessing methods enforce dimensionality checks with fallback expansions, preventing shape mismatches during prediction. The Riemannian implementation additionally guards `predict_proba()` calls with `AttributeError` handling, surfacing clear error messages when probability estimation is unsupported, and validates pipeline initialization before every prediction call to prevent cryptic failures from unloaded models. This layered validation ensures inference operations fail early with actionable feedback rather than producing silent errors or corrupted outputs.

Main

Unified training orchestration

All three classification models share a common `main.py` entry point structure that orchestrates the complete training pipeline from data loading through model persistence. Each script accepts command-line arguments specifying analysis type (ERP, time–frequency, spectral, connectivity, inter-trial coherence), data folder path, and optional JSON-encoded class filters for subset training. Configuration management follows a consistent pattern: loading analysis-specific JSON files from a local `configs` directory, merging with hardcoded defaults, and extracting hyperparameters (learning rates, batch sizes, epochs, regularization parameters). The unified workflow proceeds through five stages: configuration parsing, data loading via `PreprocessBridge`, subject-wise data splitting or cross-validation, model initialization and training, and weight persistence with standardized naming conventions. All scripts suppress framework warnings (TensorFlow logging, NumPy deprecations) to maintain clean console output for UI integration.

Common training logic

Shared responsibilities span data preparation, label construction, and evaluation reporting. All implementations: (i) invoke `PreprocessBridge.load_and_transform()` with the appropriate analysis key and model target, receiving `X` tensors and `y` metadata dictionaries containing condition (target/standard/novelty), group (HC/PD), and subject IDs; (ii) apply optional class filtering when `selected_classes` arguments specify diagnostic-condition subsets (e.g., PD-target only), re-indexing samples while preserving subject associations; (iii) construct combined multi-class labels via `groups * 3 + conditions`, then remap to contiguous ranges matching the filtered class count to avoid sparse label spaces; (iv) perform subject-wise splits to prevent data leakage, ensuring no subject appears in multiple partitions—typical ratios are 60% training, 20% validation, 20% test, with GroupKFold for robust estimates; (v) handle edge cases (fewer than three subjects, empty validation sets) by printing warnings and adjusting split logic; (vi) save best models using descriptive filenames encoding analysis type and filtered groups (e.g., `EEGNet_erp_gPD_weights_best.keras`), placing artifacts directly in the data folder without creating separate results hierarchies; and (vii) report training, validation, and test accuracies plus loss metrics in structured console blocks for upstream JSON parsing.

Deep learning workflows (EEGNet and EEG-Inception)

Neural network training scripts leverage TensorFlow/Keras APIs with categorical cross-entropy loss and Adam optimization. Shared elements include: one-hot encoding labels via `to_categorical()` after subject splits; initializing models with architecture-specific hyperparameters (filter counts, kernel sizes, dropout rates) extracted from configs; compiling with `model.compile()` specifying loss, optimizer, and accuracy metrics; training via `model.fit()` with validation data, epoch counts, and callback arrays; using `ModelCheckpoint` to monitor validation accuracy and persist only the best-performing weights; and loading saved weights post-training for final test-set evaluation. Both scripts explicitly set `K.set_image_data_format('channels_l` to enforce Keras' channel-last convention, matching preprocessing outputs. EEGNet applies a simple transpose `(batch, 1, channels, time) -> (batch, channels, time, 1)` to align bridge outputs with its depthwise-separable architecture. EEG-Inception implements more complex reshaping: detecting whether inputs are 3D (ERP: channels, time) or 4D (time–frequency/inter-trial: frequency, time, channels), applying model-specific transpositions to achieve `(batch, time, channels, 1)` layouts, and selecting between `EEGInception` (for ERP) and `EEGInceptionTF` (for spectrograms) model constructors based on analysis type. EEG-Inception additionally supports optional data augmentation for time–frequency inputs (noise injection, time/frequency masking, amplitude scaling, mixup), controlled via config flags, to combat overfitting with limited samples.

Riemannian workflow (pyRiemann and scikit-learn)

The Riemannian training script diverges significantly by using traditional machine learning pipelines instead of gradient descent. Key differences: (i) imputes NaN/Inf values with zeros before covariance estimation to avoid singular matrices; (ii) invokes model factories (`create_riemannian_pipeline()`, `create_connectivity_pipeline()`) that return complete `sklearn.pipeline.Pipeline` objects encapsulating covariance estimators (SCM, LWF with shrinkage), tangent space projections, dimensionality reduction, and SVM/MDM classifiers; (iii) supports deep tangent space networks when `use_deep=True`, injecting custom neural estimators with configurable hidden dimensions, dropout, and PyTorch Skorch wrappers into the pipeline; (iv) performs five-fold `GroupKFold` cross-validation on the full dataset to report

mean and standard deviation accuracies, providing robust performance estimates despite small sample sizes; (v) trains a final model on the complete dataset after CV, explicitly warning users that full-data training accuracy is optimistic and CV metrics represent true generalization; (vi) persists the entire pipeline (including fitted transformers and classifiers) via `joblib.dump()` to `.pkl` files; and (vii) prints detailed label summaries showing group/condition/combined-class distributions plus subject counts, aiding debugging of aggressive filtering that might leave too few samples. Connectivity analysis uses regularized covariance estimators with configurable shrinkage and minimum eigenvalue thresholds to stabilize ill-conditioned matrices, while spectral analysis operates on pre-averaged frequency-band power.

Configuration and hyperparameter management

All scripts define analysis-specific config files (`erp_config.json`, `tf_config.json`, `spectral_config.j` `connectivity_config.json`) mapping user-facing analysis modes to backend parameter sets. Configs specify model architecture (layer counts, filter dimensions), training regimes (epochs, batch sizes, learning rates), regularization (dropout, weight decay, label smoothing), optimization strategies (schedulers, early stopping patience), and workflow toggles (enable K-fold, grid search, augmentation). Scripts merge loaded configs with defaults via dictionary updates, allowing partial overrides. EEG-Inception exposes advanced options like target validation accuracy for early stopping (`StopOnAccuracy` callback) and learning rate reduction on plateaus. Riemannian configs include traditional ML parameters (SVM kernels, regularization constants, class weights) plus deep tangent network settings when enabled. This externalized configuration supports rapid experimentation without code modification, though all scripts also accept command-line overrides for data paths and class filters, enabling controller-driven batch training across multiple analysis-class combinations.

Subject-wise splitting and cross-validation

Preventing subject leakage is critical: all scripts partition data by unique subject IDs rather than random trial shuffling. EEGNet and EEG-Inception default to fixed 60/20/20 train/validation/test splits using `train_test_split()` with `random_state=42` for reproducibility, extracting subject lists first then masking samples via `np.isin(subject_ids, train_subs)`. When fewer

than three subjects exist, scripts issue warnings and allocate all data to training. EEG-Inception optionally supports `StratifiedGroupKFold` when `use_kfold=True`, iterating folds, training separate models, and reporting per-fold accuracies plus mean/std aggregates before final training on the full set. Riemannian always uses five-fold `GroupKFold`, treating each fold independently without final model averaging, then trains one last pipeline on all data for deployment. This cross-validation rigor addresses the Parkinson's dataset's limited subject count (approximately 20 HC, 20 PD), ensuring reported accuracies reflect out-of-subject generalization rather than memorization.

Output and logging conventions

All scripts print structured console messages designed for upstream parsing by the classification controller. Key patterns: section headers delimited by repeated dashes or equals signs; accuracy/loss metrics formatted as percentages with fixed decimal places; explicit labels for train/validation/test partitions; and warnings prefixed with "Warning:" or "Error:" for UI highlighting. None generate standalone results folders or plots—training completion messages specify only the saved weight path. This minimalist output philosophy keeps the data folder clean and delegates visualization to the controller or separate analysis tools. Errors are surfaced via exceptions that propagate to the controller's subprocess capture, where traceback parsing extracts actionable messages for the user. The combination of clean logs and standardized weight naming enables the controller to discover trained models programmatically and populate UI dropdown lists without manual intervention.

Test

Unified evaluation architecture

All three classification models provide standalone `test.py` scripts that independently evaluate trained models on held-out test sets, reporting aggregate metrics and per-class performance without requiring the full training pipeline. Each script accepts command-line arguments specifying analysis type, data folder path, and optional pre-trained weight file paths. When weight paths are omitted, scripts auto-detect model files using standardized naming conventions (e.g., `EEGNet_erp_weights_best.keras`). The evaluation workflow mirrors training's data load-

ing and splitting phases to ensure test subjects remain isolated from training, then loads saved model artifacts, performs inference on test samples, and outputs structured JSON payloads tagged with `JSON_RESULT:` prefixes for controller parsing. This separation allows post-hoc model evaluation without re-training and supports interactive single-sample classification via weight reuse.

Common evaluation logic

Shared responsibilities span data loading, split reproduction, balanced sampling, and result reporting. All implementations: (i) initialize `PreprocessBridge` and invoke `load_and_transform()` with matching analysis types and model targets used during training; (ii) reproduce identical subject-wise train/validation/test splits by applying `train_test_split()` with `random_state=42`, ensuring test subjects never appeared in training data—this split fidelity is critical for unbiased accuracy estimates; (iii) construct combined multi-class labels via the same `groups * 3 + conditions` formula, maintaining label consistency across training and testing phases; (iv) apply automatic balanced sampling by detecting minimum sample counts across all unique labels and downsampling majority classes to match, preventing class imbalance from inflating accuracy metrics—each class contributes equally to reported performance; (v) load pre-trained model artifacts (Keras `.keras` files or `joblib` `.pkl` pipelines) with defensive existence checks and graceful error messages; (vi) compute overall test accuracy plus per-class accuracies, iterating through unique labels and calculating class-specific correct prediction rates; (vii) format console output with delimited result blocks showing test accuracy, per-class breakdowns, sample counts, and subject counts; and (viii) emit JSON-encoded result dictionaries containing numeric metrics, class names, sample/subject counts, and dataset identifiers for UI consumption. The `JSON_RESULT:` prefix enables controller regex matching to extract structured data from subprocess stdout streams.

Deep learning evaluation (EEGNet and EEG-Inception)

Neural network test scripts leverage TensorFlow/Keras model loading and prediction APIs. Shared elements include: setting `K.set_image_data_format('channels_last')` for tensor layout consistency; loading complete model graphs with `load_model()` from `.keras` files

(Keras 3.x format); extracting label metadata (conditions, groups, subject IDs, dataset names) from bridge outputs; filtering invalid groups and conditions with boolean masks to handle corrupted or out-of-range labels, printing warnings for dropped samples; applying identical tensor reshaping as training—EEGNet transposes to `(batch, channels, time, 1)`, EEG-Inception performs dimension detection and reordering to achieve `(batch, time/frequency, time, channels)` layouts based on analysis type; one-hot encoding test labels via `to_categorical()` to match model output shapes; invoking `model.evaluate()` for aggregate accuracy and loss, then `model.predict()` for per-sample class probabilities; extracting predicted classes via `argmax()` over the probability axis; and computing per-class accuracy by masking test samples belonging to each class and calculating correct prediction rates. EEG-Inception additionally implements dynamic class detection by inspecting `model.output_shape[-1]` to infer whether the model was trained for binary (HC vs PD), tri-class (conditions only), or six-class (group × condition) classification, adjusting label construction and class names accordingly. Both scripts build detailed per-sample result arrays containing predicted/actual labels, confidence scores (max probability), correctness flags, and full probability distributions across all classes—EEG-Inception exposes this granular data in its JSON output to support UI-driven sample inspection and confusion matrix generation.

Riemannian evaluation (pyRiemann pipeline)

The Riemannian test script diverges by loading complete `scikit-learn` pipelines rather than model weights alone. Key differences: (i) uses `joblib.load()` to deserialize `.pkl` files containing fitted covariance estimators, tangent space projectors, and SVM/MDM classifiers; (ii) skips one-hot encoding since sklearn classifiers accept integer labels directly; (iii) applies NaN/Inf imputation (`np.nan_to_num()`) before evaluation to prevent covariance estimation failures on missing values; (iv) invokes `clf.score()` for overall accuracy rather than separate loss metrics (no loss functions in traditional ML); (v) extracts predictions via `clf.predict()` without explicit probability retrieval, as MDM classifiers lack `predict_proba()` methods; (vi) computes per-class accuracies by iterating unique condition labels (target/standard/novelty) rather than combined group-condition classes, reflecting Riemannian's focus on spectral/connectivity features that may not encode group differences as robustly; (vii) includes dataset names in

JSON output but omits per-sample granular results (no probability distributions); and (viii) prints simpler console summaries without loss values. The pipeline approach ensures all preprocessing transformations (covariance, tangent projection) are applied consistently between training and testing without manual parameter alignment.

Balanced sampling and class distribution

All scripts implement automatic class balancing to address inherent dataset imbalance (e.g., more standard trials than target trials, unequal HC/PD subject counts). The algorithm: (i) identifies unique labels in the test set via `np.unique()`; (ii) counts samples per label using `np.sum(labels == label)`; (iii) finds the minimum count across all labels; (iv) for each label, randomly samples exactly `min_samples_per_label` trials without replacement via `np.random.choice()`, ensuring no class dominates; (v) concatenates sampled indices, shuffles them to mix labels, and extracts corresponding `X_test` and `y_test` subsets; and (vi) prints per-class sample counts to confirm balance. This approach prevents scenarios where a model achieves high accuracy by simply predicting the majority class—balanced evaluation forces models to perform equally well across all diagnostic groups and stimulus conditions. EEGNet and EEG-Inception apply balancing to combined six-class labels (group × condition), while Riemannian balances three-class condition labels. The `random_state` is *not* fixed for sampling (uses default random seed), introducing minor variability across runs but maintaining fairness.

Split reproduction and leakage prevention

Critical to unbiased evaluation is reproducing training's exact subject-wise splits using identical random seeds. All scripts invoke `train_test_split()` twice with `random_state=42`: first splitting 60% train vs 40% temp, then splitting temp into 20% validation vs 20% test. This matches training's logic exactly, ensuring test subjects `test_subs` are identical between training and testing runs given the same input data. Masking via `np.isin(subject_ids, test_subs)` isolates test samples before any processing, guaranteeing no subject leakage. Scripts validate that `len(unique_subjects) >= 3` before splitting; fewer subjects trigger fallback logic allocating all data to test with printed warnings. This rigor prevents optimistic

accuracy inflation from subject memorization—models never see test subjects' data during training, forcing genuine generalization across individuals rather than trial-level overfitting.

Output formatting and JSON serialization

All scripts follow strict output conventions for controller integration. Console output uses repeated equals signs (`=*60`) to delimit result blocks, labels metrics explicitly (`Test Accuracy:`, `Per-Class Accuracy:`), and formats percentages with two decimal places. JSON payloads encode: `test_accuracy` (float), `class_accuracies` (dict mapping class names to floats), `num_test_samples`/`num_test_subjects` (ints), and `dataset_names` (list of strings identifying trial sources). Deep learning scripts add `test_loss`, `per_sample_results` (detailed prediction arrays), and `classification_mode` (e.g., "6-class"). EEGNet includes `group_counts` and `subject_group_labels` mapping subject IDs to diagnostic groups. All floats are cast explicitly via `float()`/`int()` to avoid NumPy type serialization errors. EEG-Inception implements `sanitize_for_json()` to replace NaN/Inf with `None`, preventing `json.dumps()` failures. The `ensure_ascii=True` flag escapes non-ASCII characters in dataset paths. The `JSON_RESULT:` prefix enables regex extraction: `re.search(r'JSON_RESULT:(.*)', stdout)` captures the payload for UI parsing without manual text processing.

II.7 Limitations

II.8 Summary

Chapter III

Classification Results

Classification combinations that belong to an analysis were evaluated using the same hyper-parameters, therefore the results may vary regarding accuracy and loss. These can be tuned further for better individual accuracies.

III.1 EEGNet Classification Results

III.1.1 ERP Data Classification Results

The hyperparameters used were:

- `num_classes`: 3
- `batch_size`: 16
- `epochs`: 50
- `lr`: 0.001
- `dropout`: 0.5
- `f1`: 8
- `d`: 2
- `f2`: 16
- `kern_length`: 64

Target Control Group vs Parkinson's Disease

Table III.1: EEGNet ERP classification performance (target vs Parkinson's Disease)

Split	Accuracy (%)	Loss
Training	63.33	0.6884
Validation	60.00	0.6972
Test	50.00	0.6921

Standard Control Group vs Parkinson's Disease

Table III.2: EEGNet ERP classification performance (standard vs Parkinson's Disease)

Split	Accuracy (%)	Loss
Training	86.67	0.5161
Validation	80.00	0.5632
Test	70.00	0.7055

Novelty Control Group vs Parkinson's Disease

Table III.3: EEGNet ERP classification performance (novelty vs Parkinson's Disease)

Split	Accuracy (%)	Loss
Training	56.67	0.6900
Validation	80.00	0.6760
Test	40.00	0.7043

III.2 EEG-Inception Classification Results

III.2.1 Time-Frequency Data Classification Results

Here are the hyperpatameters used for the time-frequency classification:

- `batch_size`: 12
- `epochs`: 200
- `learning_rate`: 0.002

- dropout_rate: 0.25
- filters_per_branch: 12
- scales_time: [6000, 4000, 2000]
- activation: elu
- fs: 1000
- normalize_data: true
- use_kfold: true
- n_folds: 5

Target Control Group vs Parkinson's Disease

Table III.4: EEG-Inception Time-Frequency K-Fold Cross-Validation Results (Target vs Parkinson's Disease)

Fold	**Accuracy (%)**
Fold 1	50.0
Fold 2	60.0
Fold 3	70.0
Fold 4	80.0
Fold 5	70.0
Mean	66.00 ± 10.20
Min	50.00
Max	80.00

Table III.5: Classification Report for Best Model (Accuracy: 80.00%)

Class	**Precision**	**Recall**	**F1-Score**
HC-target	0.61	0.88	0.72
PD-target	0.79	0.44	0.56

Standard Control Group vs Parkinson's Disease

Table III.6: EEG-Inception Time-Frequency K-Fold Cross-Validation Results (Standard vs Parkinson's Disease)

Fold	**Accuracy (%)**
Fold 1	50.0
Fold 2	60.0
Fold 3	70.0
Fold 4	70.0
Fold 5	70.0
Mean	64.00 ± 8.00
Min	50.00
Max	70.00

Table III.7: Classification Report for Best Model (Accuracy: 70.00%)

Class	**Precision**	**Recall**	**F1-Score**
HC-standard	0.59	0.88	0.71
PD-standard	0.77	0.40	0.53

Novelty Control Group vs Parkinson's Disease

Table III.8: EEG-Inception Time-Frequency K-Fold Cross-Validation Results (Novelty vs Parkinson's Disease)

Fold	**Accuracy (%)**
Fold 1	70.0
Fold 2	70.0
Fold 3	50.0
Fold 4	70.0
Fold 5	80.0
Mean	68.00 ± 9.80
Min	50.00
Max	80.00

Table III.9: Classification Report for Best Model (Accuracy: 80.00%)

Class	**Precision**	**Recall**	**F1-Score**
HC-novelty	0.71	0.60	0.65
PD-novelty	0.66	0.76	0.70

III.2.2 Inter-Trial Coherence Classification Results

Here are the hyperpatameters used for the inter-trial coherence classification:

- `batch_size`: 12
- `epochs`: 50
- `learning_rate`: 0.002
- `dropout_rate`: 0.25
- `filters_per_branch`: 12
- `scales_time`: [500, 250, 125]
- `activation`: elu
- `fs`: 401
- `use_kfold`: true
- `n_folds`: 5

Target Control Group vs Parkinson's Disease

Table III.10: EEG-Inception Inter-Trial Coherence K-Fold Cross-Validation Results (Target vs Parkinson's Disease)

Fold	**Accuracy (%)**
Fold 1	50.0
Fold 2	50.0
Fold 3	50.0
Fold 4	70.0
Fold 5	60.0
Mean	56.00 ± 8.00
Min	50.00
Max	70.00

Table III.11: Classification Report for Best Model (Accuracy: 70.00%)

Class	Precision	Recall	F1-Score
HC-target	0.57	0.48	0.52
PD-target	0.55	0.64	0.59

Table III.12: EEG-Inception Inter-Trial Coherence K-Fold Cross-Validation Results (Standard vs Parkinson's Disease)

Fold	Accuracy (%)
Fold 1	50.0
Fold 2	50.0
Fold 3	50.0
Fold 4	50.0
Fold 5	50.0
Mean	50.00 ± 0.00
Min	50.00
Max	50.00

Standard Control Group vs Parkinson's Disease

Novelty Control Group vs Parkinson's Disease

III.3 Riemannian Classification Results

III.3.1 Spectral Classification Results

Here are the hyperpatameters used for the spectral classification:

- `estimator`: oas
- `metric`: riemann
- `kernel`: linear
- `C`: 0.05
- `use_mdm`: false
- `use_deep`: true
- `enable_grid_search`: false

Table III.13: Classification Report for Best Model (Accuracy: 50.00%)

Class	Precision	Recall	F1-Score
HC-standard	0.50	0.60	0.55
PD-standard	0.50	0.40	0.44

Table III.14: EEG-Inception Inter-Trial Coherence K-Fold Cross-Validation Results (Novelty vs Parkinson's Disease)

Fold	Accuracy (%)
Fold 1	50.0
Fold 2	50.0
Fold 3	70.0
Fold 4	50.0
Fold 5	60.0
Mean	56.00 ± 8.00
Min	50.00
Max	70.00

- `test_size`: 0.8
- `val_split`: 0.6
- `random_state`: 42
- `hidden_dims`: [64, 32]
- `dropout`: 0.65
- `input_dropout`: 0.2
- `lr`: 0.0003
- `max_epochs`: 50
- `batch_size`: 128
- `weight_decay`: 0.001
- `patience`: 7
- `train_split`: 0.25
- `label_smoothing`: 0.1

Table III.15: Classification Report for Best Model (Accuracy: 70.00%)

Class	Precision	Recall	F1-Score
HC-novelty	0.59	0.40	0.48
PD-novelty	0.55	0.72	0.62

Target Control Group vs Parkinson's Disease

Table III.16: Riemannian Spectral K-Fold Cross-Validation Results (Target vs Parkinson's Disease)

Fold	Accuracy (%)
Fold 1	60.0
Fold 2	60.0
Fold 3	70.0
Fold 4	60.0
Fold 5	40.0
Mean	58.00 ± 9.80
Min	40.00
Max	70.00

Standard Control Group vs Parkinson's Disease

Table III.17: Riemannian Spectral K-Fold Cross-Validation Results (Standard vs Parkinson's Disease)

Fold	Accuracy (%)
Fold 1	70.0
Fold 2	40.0
Fold 3	20.0
Fold 4	30.0
Fold 5	30.0
Mean	38.00 ± 17.20
Min	20.00
Max	70.00

Table III.18: Riemannian Spectral K-Fold Cross-Validation Results (Novelty vs Parkinson's Disease)

Fold	Accuracy (%)
Fold 1	30.0
Fold 2	80.0
Fold 3	30.0
Fold 4	70.0
Fold 5	60.0
Mean	54.00 ± 20.59
Min	30.00
Max	80.00

Table III.19: Riemannian Channelwise Connectivity K-Fold Cross-Validation Results (Target vs Parkinson's Disease)

Fold	Accuracy (%)
Fold 1	50.0
Fold 2	50.0
Fold 3	70.0
Fold 4	60.0
Fold 5	40.0
Mean	54.00 ± 10.20
Min	40.00
Max	70.00

Novelty Control Group vs Parkinson's Disease

III.3.2 Channelwise Connectivity Classification Results

Target Control Group vs Parkinson's Disease

Standard Control Group vs Parkinson's Disease

Table III.20: Riemannian Channelwise Connectivity K-Fold Cross-Validation Results (Standard vs Parkinson's Disease)

Fold	Accuracy (%)
Fold 1	70.0
Fold 2	20.0
Fold 3	40.0
Fold 4	40.0
Fold 5	60.0
Mean	46.00 ± 17.44
Min	20.00
Max	70.00

Novelty Control Group vs Parkinson's Disease

Table III.21: Riemannian Channelwise Connectivity K-Fold Cross-Validation Results (Novelty vs Parkinson's Disease)

Fold	**Accuracy (%)**
Fold 1	80.0
Fold 2	30.0
Fold 3	60.0
Fold 4	50.0
Fold 5	60.0
Mean	56.00 ± 16.25
Min	30.00
Max	80.00

Chapter IV

Conclusion and Future Work

IV.1 Conclusion

IV.2 Future Work

Appendix A

Assessment Activity

(Content...)

Appendix B

Assessment Rubric

(Content...)